



**VIDYA VIKAS INSTITUTE OF
ENGINEERING AND TECHNOLOGY,
MYSORE**

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VTU Previous Question Papers

“Civil Engineering”

**III, V & VII Semester
Dec.2024/Jan.2025**

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VVIET**

Bannur road, Alanahalli, Mysore-570028

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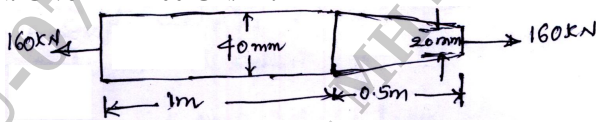
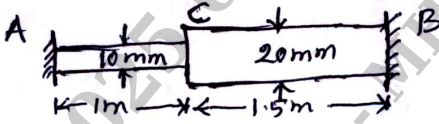
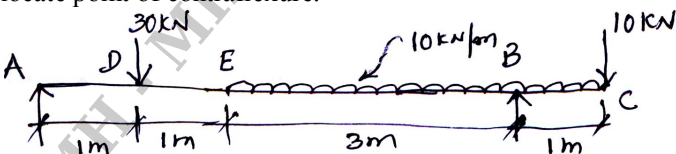
Third Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Strength of Materials

Time: 3 hrs.

Max. Marks: 100

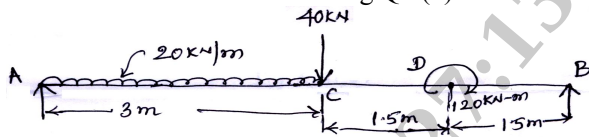
Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. *M* : Marks , *L*: Bloom's level , *C*: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Sketch a typical stress-strain curve for a ductile material and explain briefly the salient features of the curve.	6	L2	CO1
	b.	Define: i) Strain ii) Poisson's ratio.	4	L1	CO1
	c.	A 1.5 m long steel bar is having uniform diameter of 40 mm for a length of 1 m. In the next 0.5 m its diameter gradually reduces from 40 mm to 20 mm as shown in Fig.Q.1(c). Determine the elongation of this bar when subjected to an axial tensile load of 160 kN. Given $E = 200 \text{ GN/m}^2$.	10	L3	CO1
 Fig.Q.1(c)					
OR					
Q.2	a.	What are the elastic constants and explain them briefly?	6	L2	CO1
	b.	Explain St Venant's principle.	4	L2	CO1
	c.	The steel rod shown in Fig.Q.2(c) is in two parts. It has a diameter of 10 mm for a length of 1 m and 20 mm for the remaining length of 1.5 m. If it is constrained between two parts A and B and is stress free at 20°C. Find the stress in the material, when it is subjected to 70°C. $E = 200 \text{ GPa}$, $\alpha = 12 \times 10^{-6}/^\circ\text{C}$.	10	L3	CO1
 Fig.Q.2(c)					
Module – 2					
Q.3	a.	For a cantilever beam subjected to a udl of intensity w /unit length throughout, plot SFD and BMD.	6	L3	CO2
	b.	For a simply supported overhanging beam as shown in Fig.Q.3(b), draw SFD and BMD and locate point of contraflexure.	14	L3	CO2
 Fig.Q.3(b)					

1 of 3

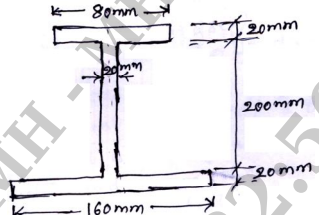
OR

Q.4	a.	For a simply supported beam subjected to udl of w /unit length throughout. Plot SFD and BMD.	6	L3	CO2
	b.	Draw SFD and BMD for the beam shown in Fig.Q.4(b).  Fig.Q.4(b)	14	L3	CO2

Module – 3

Q.5	a.	Derive the relation between bending stress and radius of curvature $\frac{\sigma}{y} = \frac{E}{R}$.	6	L3	CO3
	b.	Define: i) Neutral axis ii) Section modulus	4	L2	CO3
	c.	A hollow propeller shaft of a steam ship is to transmit 3750 KW at 240 rpm. If the internal diameter is 0.8 times the external diameter and if the maximum shear stress developed is to be limited to 160 N/mm ² , determine the size of the shaft.	10	L3	CO4

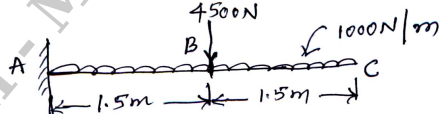
OR

Q.6	a.	Derive the torsion equation with usual notations.	8	L3	CO4
	b.	The unsymmetrical I-section shown in Fig.Q.6(b) is subjected to a shear force of 40 KN. Draw the shear stress variation diagram across the depth.  Fig.Q.6(b)	12	L3	CO3

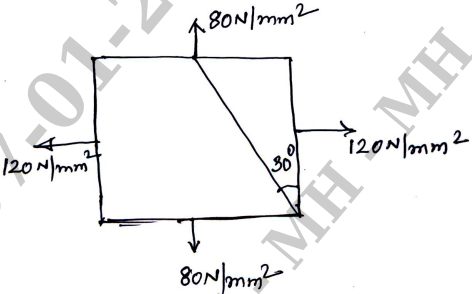
Module – 4

Q.7	a.	Derive differential equation for beam deflection with usual notations.	8	L3	CO4
	b.	Determine the critical load for a hollow cast iron rectangular column of external dimensions 200 mm × 150 mm with the thickness of the metal being 25 mm. The height of the column is 6 m and both ends are fixed. Use Euler's formula and compare the value with that obtained by using Rankine's formula taking $\sigma_c = 500$ N/mm ² and $a = 1/1600$, which of the above formula decides the safe crippling load?	12	L3	CO4

OR

Q.8	a.	Derive an expression for Euler's crippling load for both ends hinged columns with usual notations.	8	L3	CO4
	b.	Compute the mid-span and maximum deflection for the beam shown in Fig.Q.8(b). Given $E = 210$ GN/m ² and moment of inertia = 36000×10^{-9} m ⁴ .  Fig.Q.8(b)	12	L3	CO4

Module – 5

Q.9	a.	With a neat sketch, explain the two-dimensional stress system.	8	L2	CO5
	b.	A cylindrical shell is 3 m long and is having 1 m internal diameter and 15 mm thickness. Calculate the maximum intensity of shear stress induced and also the changes in the dimensions of the shell, if it is subjected to an internal fluid pressure of 1.5 N/mm^2 .	12	L3	CO5
OR					
Q.10	a.	Derive Lamé's equation for radial and hoop stress for thick cylinder subjected to internal and external fluid pressure.	10	L3	CO5
	b.	The direct stresses acting at a point in a strained material are shown in Fig.Q.10(b). Find the normal, tangential and resultant stresses on plane 30° to the plane of major principal stress. Find the obliquity of the resultant stress also.  Fig.Q.10(b)	10	L3	CO5

CBCS SCHEME

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BCV302

Third Semester B.E/B.Tech. Degree Examination, Dec.2024/Jan.2025 Engineering Survey

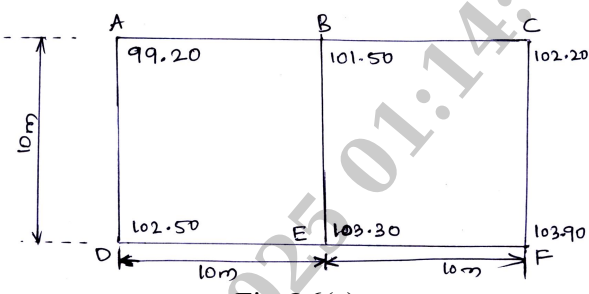
Time: 3 hrs.

Max. Marks:100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
1	a.	What is Surveying? Explain the principles of surveying.	8	L2	CO1
	b.	Compare between : i) True and magnetic bearing ii) Plane and geodetic surveying iii) Cadastral and hydrographic surveying.	6	L1	CO1
	c.	Write the advantages and disadvantages of plane table surveying.	6	L1	CO1
OR					
2	a.	Discuss on importance of surveying in civil engineering.	6	L2	CO1
	b.	Explain the different methods of measuring distances with examples.	6	L2	CO1
	c.	Differentiate between : i) Chain and chainage ii) Laser distance meter and distance measuring wheel iii) EDM and GPS iv) Topographical and construction survey.	8	L1	CO1
Module – 2					
3	a.	The following consecutive readings were taken with a level and a 4m staff on a continuously sloping ground at a common interval of 20m : 0.780, 1.535, 1.955, 2.430, 2.985, 3.480, 1.155, 1.960, 2.365, 3.640, 0.935, 1.045, 1.630 and 2.545. The RL of first point A was 180.750m. Rule out a page of level field book and enter the above readings. Compute the RL's by HI method. Also calculate the gradient of the line joining the first and last points.	10	L3	CO2
	b.	List the salient features of total station.	6	L2	CO2
	c.	Why fly and flyback leveling is required?	4	L2	CO2
OR					
4	a.	Explain the temporary adjustments of dumpy level.	6	L2	CO2
	b.	Explain the method of measuring horizontal angle by repetition method along with the tabular column.	10	L3	CO2
	c.	Discuss on different fundamental measurements of total station.	4	L1	CO2
Module – 3					
5	a.	What are contours? Explain the characteristics of contours with neat sketches.	10	L3	CO3
	b.	Brief on longitudinal and cross-sectioning with typical sketches.	8	L2	CO3
	c.	What are the input data required while creating job file in total station?	2	L1	CO3
1 of 2					

OR

6	a.	Plot the contours of RL 100.00, 101.00, 102.00, and 103.00 in the given square blocks of 10 m × 10 m. The reduced levels of guide points are given in Fig.Q6(a).	10	L3	CO3
		 Fig.Q6(a)			
	b.	Explain the following related to total station : i) Back sight data ii) Coordinates data iii) Command to plot contours in auto CAD. iv) Data transferring.	4	L2	CO3
	c.	Explain the procedure and select the contour interval.	6	L2	CO3

Module – 4

7	a.	Calculate the necessary data for setting out simple curve of radius 300m with the deflection angle of 50° 30'. The two tangents intersect at a chainage of 1192.00m. Take peg interval as 20m and tabulate the results using Rankine's method.	6	L3	CO4																		
	b.	List the different types of curves.	4	L1	CO4																		
	c.	The following perpendicular offsets were taken from chain line to an irregular boundary. Calculate the area enclosed by trapezoidal rule. <table border="1"><tr><td>Chainage (m)</td><td>0</td><td>30</td><td>60</td><td>90</td><td>120</td><td>150</td><td>180</td><td>210</td></tr><tr><td>Offset (m)</td><td>0</td><td>2.65</td><td>3.80</td><td>3.75</td><td>4.65</td><td>3.60</td><td>5.0</td><td>5.80</td></tr></table>	Chainage (m)	0	30	60	90	120	150	180	210	Offset (m)	0	2.65	3.80	3.75	4.65	3.60	5.0	5.80	10	L2	CO4
Chainage (m)	0	30	60	90	120	150	180	210															
Offset (m)	0	2.65	3.80	3.75	4.65	3.60	5.0	5.80															

OR

8	a.	A railway embankment is 10m wide with side slopes 1.5 to 1.0. Assuming ground to be level in a direction transverse to center line, calculate the volume contained in a length of 120 meters, the centre heights at 20m intervals being are 2.2, 3.7, 3.8, 4.0, 3.8, 2.8 and 2.5m. Use both trapezoidal and prismoidal method.	10	L3	CO4
	b.	Sketch out a compound curve and show the elements of it.	4	L2	CO4
	c.	Define the following related to setting out works : i) Stake ii) Post iii) Batter – board iv) Sight rail.	6	L1	CO4

Module – 5

9	a.	Discuss on the various segments of GPS.	8	L2	CO5
	b.	List the applications of RS and GIS in civil engineering.	6	L1	CO5
	c.	List out the steps in drone surveying.	6	L1	CO5

OR

10	a.	Discuss on GPS receivers.	8	L2	CO5
	b.	List the features and applications of drone surveying.	8	L1	CO5
	c.	Name the type of sensors used in drone surveying.	4	L1	CO5

CBCS SCHEME

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BCV303

Third Semester B.E/B.Tech. Degree Examination, Dec.2024/Jan.2025 Engineering Geology

Time: 3 hrs.

Max. Marks:100

**Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.**

Module – 1			M	L	C
1	a.	Explain importance of geology in Civil Engineering.	5	L2	CO1
	b.	Describe internal structure of earth with labelled sketch.	10	L2	CO1
	c.	What is land slide? Explain causes at landslides.	5	L1	CO1
OR					
2	a.	Discuss causes and effect of earthquake.	7	L2	CO2
	b.	What is Tsunami? Add a note on mitigation of Tsunami.	7	L3	CO3
	c.	Explain causes and types of volcanoes.	6	L2	CO2
Module – 2					
3	a.	What is mineral? Explain properties of mineral.	8	L2	CO2
	b.	Explain different types of igneous rocks.	6	L2	CO2
	c.	What is metamorphism? Discuss types of metamorphism.	6	L2	CO2
OR					
4	a.	Describe properties of following minerals : i) Orthoclase ii) Biotite iii) Haematite.	6	L2	CO2
	b.	Enumerate stages of development of sedimentary rock.	7	L2	CO2
	c.	Discuss important uses of different rock type.	7	L2	CO2
Module – 3					
5	a.	With a neat sketch explain soil profile.	7	L2	CO3
	b.	What is weathering? Explain types of weathering.	8	L2	CO3
	c.	Discuss classification of different soil type.	5	L2	CO3
OR					
6	a.	Explain classification of soil based on grain size.	7	L2	CO3
	b.	Discuss the effect of weathering on monumental rock.	7	L2	CO3
	c.	Write a note on : i) Drifted soil ii) Laterite soil.	6	L1	CO3
1 of 2					

Module – 4

7	a.	Explain different rock deformation and causes.	8	L2	CO4
	b.	A bed of lime stone is dip 25° East and it has width of outcrop – 160 m. Determine true and vertical thickness.	6	L3	CO4
	c.	What is fault? Explain type of fault.	6	L2	CO4

OR

8	a.	Discuss feasibility of Dam site in folded and faulted region.	7	L2	CO4
	b.	A bed of shale is dipping maximum of 45° along S 60°E. Determine the amount and apparent dip along S 70°E.	6	L2	CO4
	c.	Write a note on : i) Dip and strike ii) Out crop.	7	L2	CO4

Module – 5

9	a.	Explain different water bearing formations.	8	L2	CO5
	b.	What an Aquifer? Discuss types of aquifer.	8	L2	CO5
	c.	Write a note on co-efficient of permeability.	4	L1	CO5

OR

10	a.	Explain electrical resistivity survey method in ground water exploration.	12	L2	CO5
	b.	Discuss factors affecting on permeability of rocks.	8	L2	CO5

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Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Construction Management and Entrepreneurship

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1					M	L	C																									
Q.1	a.	Explain in detail construction project formulation.			10	L2	CO1																									
	b.	The activity data of a project is given below: <table border="1"><thead><tr><th>Activity</th><th>Preceding activity</th><th>Duration (Days)</th></tr></thead><tbody><tr><td>A</td><td>-</td><td>05</td></tr><tr><td>B</td><td>-</td><td>15</td></tr><tr><td>C</td><td>-</td><td>09</td></tr><tr><td>D</td><td>A</td><td>06</td></tr><tr><td>E</td><td>C</td><td>04</td></tr></tbody></table> Draw the network diagram, identify the critical path, project duration and free float.			Activity	Preceding activity	Duration (Days)	A	-	05	B	-	15	C	-	09	D	A	06	E	C	04	10	L3	CO1							
Activity	Preceding activity	Duration (Days)																														
A	-	05																														
B	-	15																														
C	-	09																														
D	A	06																														
E	C	04																														
OR																																
Q.2	a.	What is Work Breakdown Structure (WBS)? Mention its significance in construction project.			10	L2	CO1																									
	b.	Below given table pertains to the list of activities and their time estimates of a job: <table border="1"><thead><tr><th>Activity</th><th>Event</th><th>Optimistic time (days)</th><th>Most likely time (days)</th><th>Pessimistic time (days)</th></tr></thead><tbody><tr><td>A</td><td>1 – 2</td><td>3</td><td>7</td><td>10</td></tr><tr><td>B</td><td>1 – 3</td><td>4</td><td>8</td><td>13</td></tr><tr><td>C</td><td>2 – 4</td><td>2</td><td>2</td><td>07</td></tr><tr><td>D</td><td>3 – 4</td><td>5</td><td>8</td><td>10</td></tr></tbody></table> Draw the network and critical path. What is the expected completion time with the probability of 85%? (Take probability factor Z = 1.038)			Activity	Event	Optimistic time (days)	Most likely time (days)	Pessimistic time (days)	A	1 – 2	3	7	10	B	1 – 3	4	8	13	C	2 – 4	2	2	07	D	3 – 4	5	8	10	10	L3	CO1
Activity	Event	Optimistic time (days)	Most likely time (days)	Pessimistic time (days)																												
A	1 – 2	3	7	10																												
B	1 – 3	4	8	13																												
C	2 – 4	2	2	07																												
D	3 – 4	5	8	10																												
Module – 2																																
Q.3	a.	Discuss on Class of Labour. What are the key factors of minimum wages act 1948?			10	L2	CO2																									
	b.	List the factors affecting Labour productivity? Briefly discuss any three factors.			10	L2	CO2																									
OR																																
Q.4	a.	Enumerate the factors to be considered for selection of Construction Equipment.			10	L2	CO2																									
	b.	Explain material management and inventory management.			10	L2	CO2																									
Module – 3																																
Q.5	a.	Explain types of procurement and procurement planning.			10	L2	CO3																									
	b.	Explain the sustainable procurement management.			10	L2	CO3																									
OR																																
Q.6	a.	Explain the different types of construction contracts.			10	L2	CO3																									
	b.	Define contractor and subcontractor. Explain the effective sub contractor management.			10	L2	CO3																									

Module – 4					
Q.7	a.	Explain the process of construction project quality management.	10	L2	CO4
	b.	Explain the safety measures adopted during construction.	10	L2	CO4
OR					
Q.8	a.	Explain Safety Management and Risk Management.	10	L2	CO4
	b.	Explain the terms: i) Facilities Management ii) Occupancy certificate	10	L2	CO4
Module – 5					
Q.9	a.	Explain the different characteristics of a Successful Entrepreneur.	10	L2	CO5
	b.	Explain 5M model and communication skills.	10	L2	CO5
OR					
Q.10	a.	Explain the Business Planning process, Marketing Plan and Financial Plan.	10	L2	CO5
	b.	Explain the role and significance of venture capital.	10	L2	CO5

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BCV503

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Concrete Technology

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Use IS : 456-2000, IS 10262 : 2019 are permitted.
3. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Explain the chemical composition of ordinary Portland cement.	10	L2	CO1
	b.	Describe : i) Rapid Hardening Cement (RHC) ii) Sulphate Resisting Cement (SRC)	10	L2	CO1
OR					
Q.2	a.	Describe briefly the classification of Aggregates.	10	L2	CO1
	b.	Explain the effects of fly ash and silica fresh concrete.	10	L2	CO1
Module – 2					
Q.3	a.	Briefly explain which are the factors affecting workability.	10	L2	CO2
	b.	Briefly, explain any 2 lab test to measure the workability of fresh concrete.	10	L2	CO2
OR					
Q.4	a.	Briefly explain the necessity of compaction of concrete. Also explain different methods such as Hand Compaction and compaction by needle vibrator.	10	L2	CO2
	b.	Explain the need for curing of concrete. Also what is water curing and membrane curing?	10	L2	CO2
Module – 3					
Q.5	a.	Explain creep of concrete and factors affecting creep of concrete.	10	L2	CO3
	b.	Explain Shrinkage of concrete. Also explain plastic shrinkage and drying shrinkage of concrete.	10	L2	CO3
OR					
Q.6	a.	Describe Sulphate attack and Chloride attack on Hardened concrete.	10	L2	CO3
	b.	Explain : i) Rebound Hammer test ii) Ultrasonic pulse velocity.	10	L2	CO3
Module – 4					
Q.7		Explain Significance of concrete mix design and write the steps involved in concrete mix design as per IS code and also discuss the variables in proportioning of concrete.	20	L2	CO4

OR					
Q.8		Design a concrete mix for grade M ₂₅ a) Grade designation → M-25 b) Type of cement → OPC 53 grade c) Maximum nominal Aggregate size → 20mm d) Minimum cement content → 310 Kg/m ³ e) Maximum water cement ratio → 0.45 f) Workability → 50-75 mm (Slump) g) Exposure condition → Normal h) Degree of supervision → Good i) Type of aggregate → Crushed angular aggregate j) Maximum cement content → 540 Kg/m ³ k) Chemical admixture type → Super plasticizer l) Specific gravity of cement → 3.15 m) Specific gravity of water → 1.0 n) Specific gravity of C.A → 2.882 o) Water absorption of C.A → 1% p) Free surface moisture : Nil q) Specific gravity of fine aggregate : 2.605 r) Water absorption of fine aggregate : 1.23% s) Free surface moisture of F.A : Nil.	20	L3	CO4
Module – 5					
Q.9	a.	Explain the manufacturing process of Ready mix concrete.	10	L2	CO5
	b.	Explain the concept of Self Compacting Concrete (SCC). And its advantages and disadvantages of SCC.	10	L2	CO5
OR					
Q.10	a.	What is light weight concrete? Explain different materials used in light weight concrete.	10	L2	CO5
	b.	Explain : i) High strength concrete ii) High performance concrete.	10	L2	CO5

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CBCS SCHEME

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BCV515D

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Remote Sensing and GIS

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	What is Remote Sensing? Explain types of Remote sensing.	6	L1	CO1
	b.	Explain interaction of Electromagnetic Radiation with Earth surface.	7	L1	CO1
	c.	Discuss different types of platforms used in Remote sensing.	7	L1	CO1
OR					
Q.2	a.	Describe stages (components) of Remote sensing.	7	L2	CO1
	b.	What are sensors? Discuss Resolution of sensing.	6	L2	CO1
	c.	Explain key elements of visual interpretation.	7	L2	CO1
Module – 2					
Q.3	a.	Define photogrammetry? Explain the field of application of photogrammetry.	8	L2	CO2
	b.	Discuss advantages and disadvantages of photogrammetry.	6	L2	CO2
	c.	Write a note on Digital photogrammetry.	6	L2	CO2
OR					
Q.4	a.	What is Aerial photogrammetry? Discuss its importance.	7	L2	CO2
	b.	Write a note on : i) Geometry of vertical photogrammetry ii) Relief displacement.	7	L2	CO2
	c.	Discuss scale ground co-ordinators.	6	L2	CO2
Module – 3					
Q.5	a.	Define GIS? Explain components of GIS.	10	L2	CO3
	b.	Explain different data modules of GIS.	10	L2	CO3
OR					
Q.6	a.	Explain Functions of GIS.	8	L2	CO3
	b.	Discuss importance of Topologies in GIS.	6	L2	CO3
	c.	Enumerate Advantages/Disadvantages of Vector Data.	6	L2	CO3

Module – 4					
Q.7	a.	Explain the application of Remote Sensing (RS) in water resource management.	7	L3	CO4
	b.	Explain importance of Remote sensing in natural Resource management.	6	L3	CO4
	c.	What are the criteria considered during highway alignment using Remote sensing?	7	L3	CO4
OR					
Q.8	a.	Explain the application of RS and GIS on Accidental analysis and Traffic management.	8	L3	CO4
	b.	Write a note GIS and GPS.	6	L2	CO4
	c.	Discuss importance of GPS in Remote Sensing image interpretation.	6	L2	CO4
Module – 5					
Q.9	a.	Explain the application of Remote sensing and GIS in urban planning and Forestry.	7	L3	CO5
	b.	How Remote sensing and GIS is using in change detection study.	6	L3	CO5
	c.	Discuss application of Remote sensing and GIS in Agriculture.	7	L3	CO5
OR					
Q.10	a.	How the disaster management study analysed by using RS and GIS.	8	L3	CO5
	b.	Write a note on i) Urban sprawl ii) Circular system	6	L3	CO5
	c.	Explain the application of Remote sensing and GIS in urban area development and layout planning.	6	L3	CO5

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Question Paper Version : A

Fifth Semester B.E./B.Tech Degree Examination, Dec.2024/Jan.2025
Environmental Studies

Time: 1 hr.]

[Max. Marks: 50

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 a) Animals b) Human beings c) Plants d) Fish
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 6. The number of SDGS are
 a) 10 b) 15 c) 17 d) 16
 7. Which one of the following is not comes under SDG's?
 a) Zero Hunger b) Political activity c) Education d) Gender quality
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15. O₃ Represents
a) Oxygen b) Planets c) Ozone d) Green house gases
16. Which of the following is an example of Water Borne Diseases
a) Hepatitis - A b) Hepatitis - B c) Hepatitis - C d) Hepatitis - H
17. Identify the renewable sourced of energy from the following :
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25. Normal sound level of quiet speech is
a) 120 dB b) 90 dB c) 140 dB d) 50 dB

26. What is the permissible range of p^H for drinking water as per the Indian standard?
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37. The Ozone layer is locate upto _____ km above the earth surface.
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38. Acid rain effects on
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 a) Dental flurosis b) Bone brittling c) Dental caries d) Plague
40. Desired concentration of fluoride content in drinking water in mg/ℓ is
 a) 0.7 – 1.2 b) 0.5 – 2 c) 2 – 5 d) 1 – 1.75

41. The First United National conference as Human Environment was held at Stickholm in
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a) Effective laws b) Active participation of the publics
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a) Air b) Water c) Land d) All of these
47. Karnataka State "Pollution Control Board" was established in the year.
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48. Which of the following is not a waste treatment method for biomedical waste?
a) Incineration b) Chemical disinfecting
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49. The Organic material of the solid waste will decompose.
a) By the flow of water b) By the soil particles
c) By the action of micro organisms d) B Oxidation
50. Which of the following element make e – waste hazardous in nature?
a) Lead b) Glass c) Plastic d) Iron

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Question Paper Version : B

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Environmental Studies

Time: 1 hr.]

[Max. Marks: 50

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Question Paper Version : C

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Environmental Studies

Time: 1 hr.]

[Max. Marks: 50

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Question Paper Version : D

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Environmental Studies

Time: 1 hr.]

[Max. Marks: 50

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32. The green house gas is
a) N_2O b) CH_4 c) CO_2 d) All of these
33. Which is not a natural source of pollution?
a) Volcanoes b) Forest - Fire c) Coal - fire d) Dust storms
34. Air pollution from automobiles can be controlled by providing
a) Wet collector b) Scrubbers c) Catalytic converter d) All of these
35. Normal sound level of quiet speech is
a) 120 dB b) 90 dB c) 140 dB d) 50 dB
36. What is the permissible range of p^H for drinking water as per the Indian standard?
a) 6 to 9 b) 6.5 to 7.5 c) 6 to 8.5 d) 6.5 to 8.5
37. Which of the following is a major cause of soil pollution?
a) Accidents involving the vehicles that are transporting waste material.
b) Pesticides and chemical fertilizers from agricultural lands
c) Improper solid waste disposal d) All of these
38. What does E – Waste stand for?
a) Environment Waste b) Electronic Waste
c) Equipment Waste d) All of these
39. What is municipal solid waste disposal?
a) The placement on the land b) The lack of waste
c) In sanitary land fill site d) All of these
40. What is major cause of air pollution?
a) Fossil fuel burning b) Industry emission
c) Wild fire d) All of these
41. Solar Energy is an example of
a) Renewable b) Continuous c) Non - renewable d) None of these

42. Hydrogen energy can be tapped through
a) Heat pumps b) Fuel cells c) Petroleum d) Gassifiers
43. OTEC is an energy technology that converts
a) Energy in large tides of ocean to generate electricity
b) Energy in ocean waves to generate electricity
c) Energy in ocean due to thermal gradient to generate electricity
d) Energy in the fast moving ocean currents to generate electricity.
44. Which of the following source of energy is less eco – friendly?
a) Biogas b) Wind c) Solar d) Nuclear
45. O₃ Represents
a) Oxygen b) Planets c) Ozone d) Green house gases
46. Which of the following is an example of Water Borne Diseases
a) Hepatitis - A b) Hepatitis - B c) Hepatitis - C d) Hepatitis - H
47. Identify the renewable sourced of energy from the following :
a) Coal b) Wind power c) Uranium d) Oil
48. Which of the following is biotic?
a) Forest b) Iron c) Water d) None
49. World Environment Day celebrated on
a) 6th June b) 5th July c) 5th June d) 5th Jan
50. Which of the following is conventional source of Energy?
a) Wind Energy b) Solar Energy c) Bio gas d) None of these

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BMATEC301/BEC301/BBM301

Third Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 AV Mathematics III for EC/ BM Engineering

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Statistical table and Mathematics formula handbook are allowed.
3. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1				M	L	C																	
Q.1	a.	Obtain the Fourier series of $f(x) = \frac{\pi - x}{2}$ in $0 < x < 2\pi$. Hence deduce that $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \frac{\pi}{4}$.	6	L2	CO1																		
	b.	Find the Fourier series of $f(x) = x $ in $(-\ell, \ell)$. Hence show that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$.	7	L3	CO1																		
	c.	Expand $f(x) = 2x - 1$ as a cosine half range Fourier series in $0 < x < 1$.	7	L2	CO1																		
OR																							
Q.2	a.	Find the Fourier series of $f(x) = \begin{cases} 1 + \frac{2x}{\pi} & \text{in } -\pi < x < 0 \\ 1 - \frac{2x}{\pi} & \text{in } 0 < x < \pi \end{cases}$. Hence deduce that $\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$	6	L2	CO1																		
	b.	Obtain the sine half range series of, $f(x) = \begin{cases} \frac{1}{4} - x & \text{in } 0 < x < \frac{1}{2} \\ x - \frac{3}{4} & \text{in } \frac{1}{2} < x < 1 \end{cases}$	7	L2	CO1																		
	c.	Determine the constant term and the first cosine and sine terms of the Fourier series expansion of y from the following data : <table><tr><td>x°:</td><td>0</td><td>45</td><td>90</td><td>135</td><td>180</td><td>225</td><td>270</td><td>315</td></tr><tr><td>y:</td><td>2</td><td>$\frac{3}{2}$</td><td>1</td><td>$\frac{1}{2}$</td><td>0</td><td>$\frac{1}{2}$</td><td>1</td><td>$\frac{3}{2}$</td></tr></table>	x°:	0	45	90	135	180	225	270	315	y:	2	$\frac{3}{2}$	1	$\frac{1}{2}$	0	$\frac{1}{2}$	1	$\frac{3}{2}$	7	L1	CO1
x°:	0	45	90	135	180	225	270	315															
y:	2	$\frac{3}{2}$	1	$\frac{1}{2}$	0	$\frac{1}{2}$	1	$\frac{3}{2}$															
Module – 2																							
Q.3	a.	Find the Fourier transform of the function, $f(x) = \begin{cases} 1 & \text{for } x \leq a \\ 0 & \text{for } x > a \end{cases}$. Hence evaluate $\int_0^\infty \frac{\sin x}{x} dx$.	6	L2	CO2																		
	b.	Find the Fourier sine and cosine transforms of $f(x) = e^{-\alpha x}$, $\alpha > 0$.	7	L2	CO2																		
	c.	Find the Fourier sine transform of $\frac{e^{-ax}}{x}$, $a > 0$.	7	L3	CO2																		

OR																	
Q.4	a.	If $f(x) = \begin{cases} 1-x^2, & x < 1 \\ 0, & x \geq 1 \end{cases}$, find the Fourier transform of $f(x)$ and hence find the value of, $\int_0^{\infty} \frac{x \cos x - \sin x}{x^3} dx$.	6	L2	CO2												
	b.	Find the Fourier sine transform of $f(x) = e^{- x }$ and hence evaluate $\int_0^{\infty} \frac{x \sin mx}{1+x^2} dx$, $m > 0$.	7	L3	CO2												
	c.	Find the Discrete fast fourier of signal $= (0, 1, 49)^T$	7	L3	CO2												
Module – 3																	
Q.5	a.	Find the z-transform of, (i) $\cosh n\theta$ (ii) $\sinh n\theta$	6	L1	CO3												
	b.	If $V(z) = \frac{2z^2 + 3z + 12}{(z-1)^4}$, evaluate u_0, u_1 and u_2	7	L2	CO3												
	c.	Find the inverse z-transform of, $\frac{z}{(z-1)(z-2)}$.	7	L2	CO3												
OR																	
Q.6	a.	Solve by using z-transforms, $y_{n+2} + 2y_{n+1} + y_n = n$ with $y_0 = 0 = y_1$	6	L3	CO3												
	b.	Find $z^{-1} \left[\frac{5z}{(3z-1)(2-z)} \right]$.	7	L2	CO3												
	c.	Solve by using z-transforms $u_{n+2} - 5u_{n+1} + 6u_n = 2^n$ with $u_0 = 0 = u_1$.	7	L3	CO3												
Module – 4																	
Q.7	a.	Solve $\frac{d^3 y}{dx^3} + 6\frac{d^2 y}{dx^2} + 11\frac{dy}{dx} + 6y = 0$.	6	L1	CO4												
	b.	Solve $(D^2 + 1)y = x^2 + 4x - 6$.	7	L2	CO4												
	c.	Using the method of variation of Parameters of $\frac{d^2 y}{dx^2} - 6\frac{dy}{dx} + 9y = e^{3x}$	7	L3	CO4												
OR																	
Q.8	a.	Solve $6\frac{d^2 y}{dx^2} + 17\frac{dy}{dx} + 12y = e^{-x}$.	6	L2	CO4												
	b.	Solve the Cauchy's differential equation, $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + 8y = 65 \cos(\log x)$.	7	L2	CO4												
	c.	The charge q in a series circuit containing an Inductance L , Capacitance C , emf E satisfy the differential equation, $L \frac{d^2 q}{dt^2} + \frac{q}{C} = E$. Express q in terms of t .	7	L3	CO4												
Module – 5																	
Q.9	a.	Fit a second degree parabola $y = a + bx + cx^2$ into least square sense for the data and estimate y at $x = 6$. <table border="1" data-bbox="331 1809 670 1886"> <tr> <td>x:</td> <td>1</td> <td>2</td> <td>3</td> <td>4</td> <td>5</td> </tr> <tr> <td>y:</td> <td>10</td> <td>12</td> <td>13</td> <td>16</td> <td>19</td> </tr> </table>	x:	1	2	3	4	5	y:	10	12	13	16	19	6	L1	CO5
x:	1	2	3	4	5												
y:	10	12	13	16	19												

	b.	Find a correlation coefficient for the two variables x and y. <table><tr><td>x:</td><td>92</td><td>89</td><td>87</td><td>86</td><td>83</td><td>77</td><td>71</td><td>63</td><td>53</td><td>50</td></tr><tr><td>y:</td><td>86</td><td>83</td><td>91</td><td>77</td><td>68</td><td>85</td><td>52</td><td>82</td><td>37</td><td>57</td></tr></table>	x:	92	89	87	86	83	77	71	63	53	50	y:	86	83	91	77	68	85	52	82	37	57	7	L2	CO5
x:	92	89	87	86	83	77	71	63	53	50																	
y:	86	83	91	77	68	85	52	82	37	57																	
	c.	Ten students got the following percentage of marks in two subjects x and y. Compute the rank correlation coefficient. <table><tr><td>x:</td><td>78</td><td>36</td><td>98</td><td>25</td><td>75</td><td>82</td><td>90</td><td>62</td><td>65</td><td>39</td></tr><tr><td>y:</td><td>84</td><td>51</td><td>91</td><td>60</td><td>68</td><td>62</td><td>86</td><td>58</td><td>53</td><td>47</td></tr></table>	x:	78	36	98	25	75	82	90	62	65	39	y:	84	51	91	60	68	62	86	58	53	47	7	L2	CO5
x:	78	36	98	25	75	82	90	62	65	39																	
y:	84	51	91	60	68	62	86	58	53	47																	
OR																											
Q.10	a.	If θ is the angle between the lines of regression show that $\tan \theta = \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \left(\frac{1 - r^2}{r} \right).$	6	L2	CO5																						
	b.	Obtain the lines of regression and hence find the coefficient of correlation for the data, <table><tr><td>x:</td><td>1</td><td>3</td><td>4</td><td>2</td><td>5</td><td>8</td><td>9</td><td>10</td><td>13</td><td>15</td></tr><tr><td>y:</td><td>8</td><td>6</td><td>10</td><td>8</td><td>12</td><td>16</td><td>16</td><td>10</td><td>32</td><td>32</td></tr></table>	x:	1	3	4	2	5	8	9	10	13	15	y:	8	6	10	8	12	16	16	10	32	32	7	L2	CO5
x:	1	3	4	2	5	8	9	10	13	15																	
y:	8	6	10	8	12	16	16	10	32	32																	
	c.	If $8x - 10y + 66 = 0$ and $40x - 18y = 214$ are the two regression lines. Find $\bar{x}$, $\bar{y}$ and r. Find σ_y if $\sigma_x = 3$.	7	L2	CO5																						

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Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Research Methodology and IPR

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Identify the meaning of Research and brief out the objective and motivation in engineering research.	10	L1	CO1
	b.	Explain brief about research cycle and verify with the research flow diagram.	10	L1	CO1
OR					
Q.2	a.	Identify the types of engineering research and briefly explain them.	10	L1	CO1
	b.	Explain about the different types of research misconduct.	10	L1	CO1
Module – 2					
Q.3	a.	Explain about the importance of literature review and technical reading.	10	L2	CO2
	b.	Mention the various benefits of bibliographic databases.	10	L1	CO2
OR					
Q.4	a.	Identify the impact of technical reaction and brief about it.	10	L1	CO2
	b.	Enumerate the impact of title and keywords on citation with example.	10	L2	CO2
Module – 3					
Q.5	a.	Define Intellectual properties and explain about its types.	10	L1	CO3
	b.	Explain about the key aspect of patent law.	10	L2	CO3
OR					
Q.6	a.	Explain about the assessment of novelty.	10	L1	CO3
	b.	Brief about the patent procedure in India.	10	L1	CO4
Module – 4					
Q.7	a.	Mention and brief about the justification for copyright law.	10	L2	CO4
	b.	Explain about the basic concepts of under lying copyright law.	10	L1	CO4
OR					
Q.8	a.	Brief about the various representations of sound recordings.	10	L2	CO5
	b.	Explain about TRIPS agreement in detail.	10	L1	CO5

Module – 5					
Q.9	a.	Explain about the justification of protection designs.	10	L2	CO5
	b.	Brief about the excluded subjected matter in the context of design protection.	10	L1	CO5
OR					
Q.10	a.	What are the rights of the owner of designs? Explain.	10	L1	CO5
	b.	Brief about the Assignment of Design Rights.	10	L1	CO5

CBCS SCHEME

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Question Paper Version : A

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Environmental Studies

Time: 1 hrs.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. Environment means
 - a) A beautiful landscape
 - b) Industrial production
 - c) Sum total of all condition
 - d) None of these
 2. The term ecosystem was introduced by
 - a) Hackel
 - b) Odum
 - c) Tansley
 - d) All of these
 3. The ecological pyramid that is always upright.
 - a) Pyramid of energy
 - b) Pyramid of biomass
 - c) Pyramid of number
 - d) None of these
 4. 5th June is observed as
 - a) World Forest Day
 - b) World Environment Day
 - c) World Wildlife Day
 - d) World Population Day
 5. MoEF means
 - a) Ministry of Forest and Energy
 - b) Ministry of Environment and Forests
 - c) Ministry of Fuel and Energy
 - d) Management of Environment and Forestry
 6. In Aquatic Ecosystem Phytoplankton can be considered as a
 - a) Consumer
 - b) Producer
 - c) Macro consumer
 - d) Micro consumer
 7. _____ has Maximum Genetic Diversity in India.
 - a) Potato
 - b) Tea
 - c) Mango
 - d) Teak

8. Chipko movement was started to conserve
a) Forests b) Grasslands c) Deserts d) Soil
9. The sequence of eating and being eaten in an eco-system is called
a) Food chain b) Carbon cycle
c) Hydrological cycle d) Anthroposystem
10. Which of the following is a biotic component of an eco system?
a) Fungi b) Solar Light c) Temperature d) Humidity
11. Hydroelectric Power plant is _____
a) Non-renewable source of energy b) Conventional source of energy
c) Non-conventional source of energy d) Continuous source of energy
12. Which isotope of uranium is used for the nuclear fission reaction?
a) U-234 b) U-235 c) U-238 d) U-233
13. A solar cell is an electrical device that converts the energy of light directly into electricity by the
a) Photovoltaic effect b) Chemical effect
c) Atmospheric effect d) Physical effect
14. Bhopal gas tragedy occurred in the year
a) 1986 b) 1990 c) 1984 d) 1991
15. Wind is beneficial resource of energy as it does not cause
a) Pollution b) Echo c) Noise d) Sound
16. Which of the following is not a renewable source of energy?
a) Wind energy b) Tidal wave energy
c) Solar energy d) Fossil fuels
17. Nuclear Power Plant in Karnataka is located at
a) Bhadravathi b) Kaiga c) Sandur d) Raichur
18. In Hydro Power Plants, power is generated by
a) Hot Springs b) Wind c) Water d) Solar Energy
19. Which is not natural disaster?
a) Cyclone b) Nuclear explosion
c) Earthquake d) Volcano
20. Radiation is a health hazard because it leads to
a) Typhoid b) Cancer c) Colour blindness d) Pneumonia
21. Water pollution is caused by
a) Sewage b) Industrial effluents
c) Discharge from farms d) All of these
22. Which of the following are non-biodegradable?
a) Plastics b) Domestic sewage
c) Detergents d) Both a and c

23. Chlorine can be used
 a) To kill pathogenic micro organisms b) To increase the pH
 c) To clear the turbidity d) All of these
24. What does E-waste stand for?
 a) Environment waste b) Electronic waste
 c) Equipment waste d) None of these
25. The noise is measured in
 a) Decibels b) Joules c) PPM d) NTU
26. Maximum dissolved oxygen is required by
 a) Fish b) Bacteria c) Vegetables d) All of these
27. Colorless, odorless and non corrosive air pollutant is
 a) Sulphur dioxide b) Carbon monoxide
 c) Carbon dioxide d) Ozone
28. Which of the following is not a greenhouse gas?
 a) CO₂ b) CH₄ c) CFC d) H₂
29. For the survival of fish in a river stream, the minimum DO is prescribed
 a) 3 PPM b) 4 PPM c) 5 PPM d) 10 PPM
30. Water pollution can be minimized by
 a) Releasing sewage to ocean b) Releasing effluent to wasteland
 c) Treating wastewater d) None of these
31. Global warming could effect
 a) Climate b) Increase in sea level
 c) Melting of glaciers d) All of these
32. The primary cause of acid rain around the world is
 a) Carbon dioxide b) Sulphur dioxide
 c) Carbon monoxide d) Ozone
33. Acid rain effects on
 a) Materials b) Plants c) Soil d) All of these
34. Ozone layer is present in
 a) Troposphere b) Stratosphere
 c) Mesosphere d) Thermosphere
35. Ozone layer absorbs
 a) UV rays b) Infrared rays c) Cosmic rays d) CO
36. The Fluoride concentration for prevention of dental caries is
 a) 3 mg/L b) 2 mg/L c) 1 mg/L d) 4 mg/L
37. D.D.T is a
 a) Fungicide b) Pesticide c) Fertilizer d) Disinfectant

38. When trees are cut, amount of oxygen
a) decreases b) increases c) both a and b d) remains same
39. World ozone day is being celebrated on
a) September 5th b) October 15th c) September 16th d) September 11th
40. The effect of acid rain
a) Reduces soil fertility b) Increases atmospheric temperature
c) Skin cancer d) Causing respiratory problems
41. Which among the following is not related to GIS software?
a) CAD b) ARC GIS c) RC VIEW d) STAAD PRO
42. GIS stands for
a) Geographic Information System b) Generic Information System
c) Geological Information System d) Geographic Information System
43. Among the following _____ can be expressed as an example of hardware component?
a) Keyboard b) Arc GIS c) Autocad d) Digitalization
44. The basic requirement of any sensor system is
a) Spatial resolution b) Spectral Resolution
c) Radiometric Resolution d) All of these
45. IS 14000 standards are for the
a) Quality Management System b) Environmental Management System
c) Administration d) Supply Chain
46. What is the main objective of secondary treatment of sewage plants?
a) To remove the suspended particles
b) To remove the contaminants
c) To remove the BOD
d) To remove the organic material
47. Which of the following is the un agency on health?
a) WHO b) FAO c) UNESCO d) WTO
48. What is the full form of NGO's?
a) Non-governmental Organizations b) Non Governance Organizations
c) No Governance Organizations d) Null Governmental Organizations
49. When did green peace founded?
a) 1965 b) 1967 c) 1968 d) 1971
50. When did the Bombay Natural History Society (BNHS) founded?
a) 1857 b) 1868 c) 1883 d) 1897

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21CV51

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Hydrology and Water Resource Engineering

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain hydrologic cycle (Horton's) with process involved in it. (10 Marks)
b. What are the importance of hydrology with emphasis of global water availability. (10 Marks)

OR

- 2 a. Define rain gauge. Describe with a neat sketch the principle of working of Symon's non-recording gauge. (10 Marks)
b. A 12 – hour storm rainfall with the following depth in cm occurred over a basin :
2.0, 2.5, 7.6, 3.8, 10.6, 5.0, 7.0 10.0, 6.4, 3.8, 1.4 and 1.4.
The surface run – off resulting from the above storm is equivalent to 25.5 cm of depth over the basin. Determine the average infiltration index for the basin (10 Marks)

Module-2

- 3 a. What is runoff? List and explain Factor's affecting it. (10 Marks)
b. Define the following :
i) Basin recharge
ii) Direct run off
iii) Drainage density
iv) Over land flow. (10 Marks)

OR

- 4 a. Define "unit hydrograph" with the help of neat sketch explain various components of a flood hydrograph. (10 Marks)
b. The hourly ordinates of a two-hour unit hydrograph are given below. Derive a 6 hour unit hydrograph for the same catchment :

Time hours	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15
Discharge cumecs	0.0	1.0	2.7	5.0	8.0	9.8	9.0	7.5	6.3	5.0	4.0	2.9	2.1	1.3	0.5	0.0

(10 Marks)

Module-3

- 5 a. Discuss the benefits and ill effects of irrigation. (10 Marks)
b. Explain with a neat sketch Bondhar's irrigation. List out advantages and disadvantages. (10 Marks)

OR

- 6 a. Define duty and delta. Derive the relation between them. (08 Marks)
b. What is irrigation efficiency? Define different efficiencies of irrigation water. (06 Marks)
c. What are flow irrigation and lift irrigation. Explain briefly. (06 Marks)

Module-4

- 7 a. What is canal? List its types and explain with a neat sketch its classification based on alignment. (10 Marks)
b. What is meant by design of canal? Bring out the difference between Kennedy's and Lacey's theory. (10 Marks)

OR

- 8 a. With sketch explain different zones of a storage reservoir. (10 Marks)
b. Explain hydrological investigations of reservoir planning. List the points to be considered for selection of a site for a reservoir. (10 Marks)

Module-5

- 9 a. Explain causes of floods and what are the methods adopted to flood alleviation. (control of floods). (10 Marks)
b. Explain how flood damage analysis is estimated. (10 Marks)

OR

- 10 a. Explain the Drought? Explain the causes of drought and measures for water conservation. (10 Marks)
b. Explain the different methods of rainwater harvesting. (10 Marks)

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Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Transportation Engineering

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the characteristics of Road transport in comparison with other transport. (06 Marks)
- b. What is Jayakar Committee and what are its recommendation? How is it implemented? (06 Marks)
- c. Determine the length of different categories of roads in a state in India using the 3rd road development plan formulae and the given data :
 - i) Total area of the State = 4,30,000 Sq. km.
 - ii) Total number of towns = 550.
 - iii) Overall road Density aimed at = 75km/100 km². (08 Marks)

OR

- 2 a. List the factors affecting the Alignment. Briefly explain the Engineering surveys for Highway alignment. (06 Marks)
- b. List the silent features of: i) KSHIP ii) NHDP. (06 Marks)
- c. Fix the priority for the following system of roads assuming suitable population and productivity units : (08 Marks)

Road Link	Length (km)	Number of villages with population served			Productivity served in Tonnes	
		<1000	1001 to 2000	> 2000	Agriculture	Industrial
A	35	18	14	6	26000	10000
B	28	13	8	5	16000	12000
C	21	12	5	3	10000	14000

Module-2

- 3 a. Explain briefly any two important surface characteristics influencing highway geometric design. (06 Marks)
- b. Define Super Elevation. Derive an expression for super elevation with neat sketch. (08 Marks)
- c. Determine SSD on a : i) Level road ii) Down grade of 1 in 120.
Take $t = 2.5$ secs , $f = 0.35$, $v = 80$ kmph. (06 Marks)

OR

- 4 a. Draw a neat sketch of flexible pavement cross – section and show the component parts. Enumerate the functions of each components of the pavement. (10 Marks)
- b. Explain 'ESWL' and the graphical method of determination of ESWL for dual wheel assembly. (10 Marks)

Module-3

- 5 a. List the soil classification systems and explain briefly the HRB soil classification system. (07 Marks)
- b. What are the desirable properties of road aggregates? Mention the laboratory tests conducted to determine the property. (07 Marks)
- c. Distinguish between cutback and emulsion and list the tests on Bitumen. (06 Marks)

OR

- 6 a. Explain the procedure involved in the preparation of sub – grade layer along with quality control tests. (10 Marks)
- b. Explain step by step procedure for construction of :
i) Water Bound Macadam ii) Bituminous concrete. (10 Marks)

Module-4

- 7 a. Explain the significance and requirements of a Highway Drainage System. (10 Marks)
- b. With a neat sketch, explain lowering of water table and control of capillary rise. (10 Marks)

OR

- 8 a. Explain briefly the quantifiable and non – quantifiable benefits for a highway user. (10 Marks)
- b. Write a note on : i) BOT ii) BOOT iii) VOC. (10 Marks)

Module-5

- 9 a. In detail explain the road user characteristics. (10 Marks)
- b. Discuss the various traffic studies and what are the objectives of carrying out traffic volume studies. (10 Marks)

OR

- 10 a. Define Permanent way. Draw the neat diagram of cross – sectional view of permanent way and describe the component parts. (10 Marks)
- b. Draw a neat sketch of layout of an airport and explain the functions of each component. (10 Marks)

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CBCS SCHEME

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21CV53

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Design of RC Structural Elements

Time: 3 hrs.

Max. Marks:100

- Note:** 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Used of US 456 – 200, SP 16 is permitted
3. Missing data, if any, may be suitably assumed.

Module-1

- 1 a. Distinguish between working stresses and limit state methods design. (08 Marks)
b. Write a brief note on :
i) Balanced section
ii) Under reinforced section
iii) Deflection
iv) Cracking. (12 Marks)

OR

- 2 a. Define long-term and short-time deflection. What are the factors influencing the long-term and short-term. (08 Marks)
b. A simply supported beam of rectangular section spans over 8m and has an effective depth of 600 mm. The beam is reinforced with 1% reinforcement on the tension side. Check for the deflection control of the beam by empirical method if :
i) Fe – 250 grade steel
ii) Fe – 415 grade steel
iii) Fe – 500 grade steel. (12 Marks)

Module-2

- 3 What are the steps involved in the analysis of design, when the following area given, A_{st} or number of bars with diameter(ϕ) of bars, size of beam, grade of concrete, and steel, if load to be calculated then span is given. (20 Marks)

OR

- 4 a. Find the depth of neutral axis of a singly reinforced RC bema of 250 mm width and 500 mm effective depth. It is reinforced with 4 bars of 20 mm diameter. Use M20 concrete and FE 415 bars. Also check for type of section. (10 Marks)
b. A singly reinforced concrete beam 250 mm width is reinforced 4 bars of 25 mm diameter at an effective depth of 400 mm. If M20 grade concrete and Fe 415 bars are used, compute ultimate moment of resistance of the section. (10 Marks)

Module-3

- 5 List and enumerate general specification for flexure design of beams. (20 Marks)

OR

- 6 Design the T-beam as per IS : 456 – 2000. The beam is subjected to an ultimate moment of 400 kN-m. Use M20 concrete and Fe 415 steel. Following are the parameters which are used for design. $b_f = 800$ mm, $b_w = 200$ mm, $D_f = 100$ mm, $d = 400$ mm. (20 Marks)

Module-4

- 7 Design a simply supported slab on masonry walls to the following requirements. Draw plan and section showing reinforcement details.
Clear span = 2.5 m
Live load = 300 N/m^2
Use M-20 concrete and Fe-415 steel. (20 Marks)

OR

- 8 Design a dog-legged staircase for a building in which the vertical distance between floors is 3.6m. The stair hall measures $3\text{m} \times 6\text{m}$. Take live load on the stairs is 4 kN/m^2 . The flight are supported on 230 mm walls at the ends of outer edges. So, that it spans in the direction of going. Adopt M20 concrete and Fe-415 grade steel. Sketch the reinforcement details. (20 Marks)

Module-5

- 9 a. Distinguish between short column and long column. (10 Marks)
b. What are the assumptions made for the limit state of collapse in compression? (10 Marks)

OR

- 10 Design on RCC short square column to the following particulars.
Axial load = 1200 kN
Grade of concrete = M-20
Length of column = 1.85m
Grade of steel = Fe-415
Sketch the reinforcement details. (20 Marks)

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21CV54

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Geotechnical Engineering

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define the following with the help of three phase diagram :
i) Specific gravity
ii) Air content
iii) Degree of saturation
iv) Voids ratio
v) Water content. (10 Marks)
- b. With usual notations prove that $eS = wG$. (04 Marks)
- c. An undisturbed sample of soil has volume of 100cm^3 and mass of 100 g. On oven drying for 24 hours, the mass is reduced to 160 g, if the specific gravity of grains is 2.68. Determine the water content, voids ratio and degree of saturation of the soil. (06 Marks)

OR

- 2 a. Explain determination of in situ – density by sand replacement method. (10 Marks)
- b. Explain soil classification according to BIS classification. (06 Marks)
- c. Explain the particle size distribution curve. (04 Marks)

Module-2

- 3 a. Explain factors affecting permeability of soil. (10 Marks)
- b. With neat sketch explain constant head permeability and falling head permeability test. (10 Marks)

OR

- 4 a. Explain permeability of stratified soil deposits for both the cases. (10 Marks)
- b. Write short notes on following :
i) Effective stress
ii) Total stress
iii) Neutral stress. (06 Marks)
- c. In a falling head permeameter test, the initial head ($t = 0$) is 40 cm. The head drops by 5 cm in 10 minutes. Calculate the time required to run the test for the final head to be at 20 cm. If the sample is 6 cm in height and 50 cm^2 in cross – sectional area, Calculate the co-efficient of permeability, taking area of stand pipe is 0.5 cm^2 . (04 Marks)

Module-3

- 5 a. Explain factors affecting compaction of soil. (10 Marks)
- b. With neat sketch explain standard proctor test. (10 Marks)

OR

- 6 a. Explain mass spring analogy. (10 Marks)
b. With a neat sketch explain laboratory consolidation test. (10 Marks)

Module-4

- 7 a. With a neat sketch explain direct shear test. (10 Marks)
b. Explain Mohr – coulomb failure theory. (10 Marks)

OR

- 8 a. With a neat sketch explain Triaxial compression test. (10 Marks)
b. Explain factors affecting shear strength of soils. (05 Marks)
c. Explain Thixotrophy and sensitivity. (05 Marks)

Module-5

- 9 a. Explain the assumptions of Terzaghi's analysis for bearing capacity of soil. (07 Marks)
b. Explain effect of water table on bearing capacity of soil (08 Marks)
c. Explain factors affecting bearing capacity of soil. (05 Marks)

OR

- 10 a. Explain types of settlements. (10 Marks)
b. Estimate the immediate settlement of a concrete footing $1.5\text{m} \times 1.5\text{m}$ in size founded at a depth of 1 m in silty soil whose modulus of elasticity is 90 kg/cm^2 . The footing is expected to transmit a unit pressure of 200 kN/m^2 . (05 Marks)
c. Explain effect of eccentricity of loading for bearing capacity of soil. (05 Marks)

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Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Quantity Survey and Contract Management

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

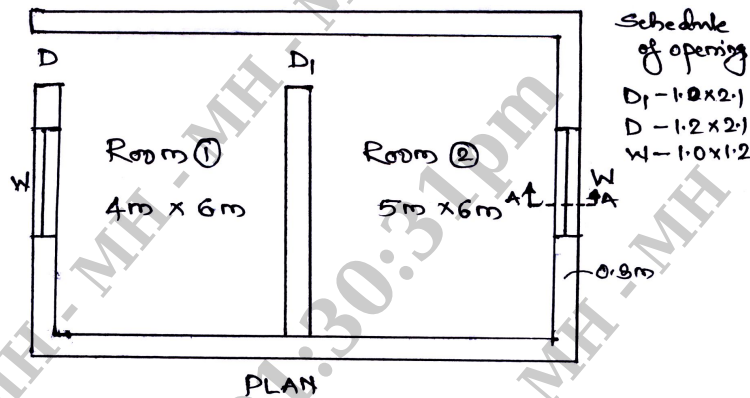
Module-1

- 1 Define estimate. Explain briefly purpose and different types of estimate (any three). (20 Marks)

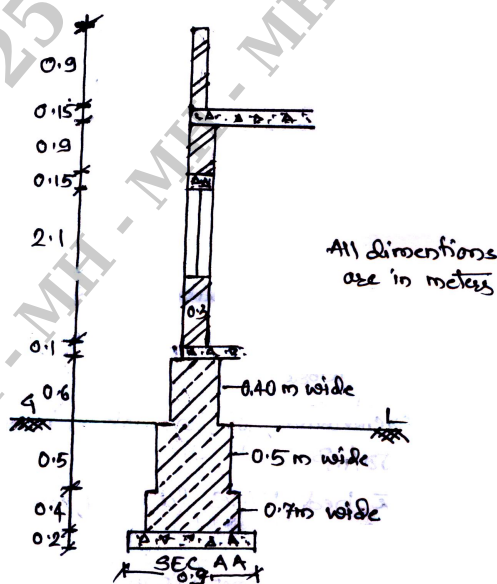
OR

- 2 The details of two room building are shown in Fig.Q.2. Estimate the quantities and cost of the following items of work:

- i) Earthwork excavation for foundation in ordinary soil @ Rs.400/m³.
- ii) Cement concrete bed 1:4:8 for wall foundation @ Rs.3800/m³.
- iii) SSM in CM 1:6 for foundation and basement @ Rs.3000/m³.
- iv) Ist class BBM work for super structure in CM 1:6 @ Rs.5800/m³.
- v) RCC 1:1.5:3 roof slab at Rs.5500/m³.



Schedule of opening
 D₁ - 1.2 x 2.1
 D - 1.2 x 2.1
 W - 1.0 x 1.2



All dimensions are in meters

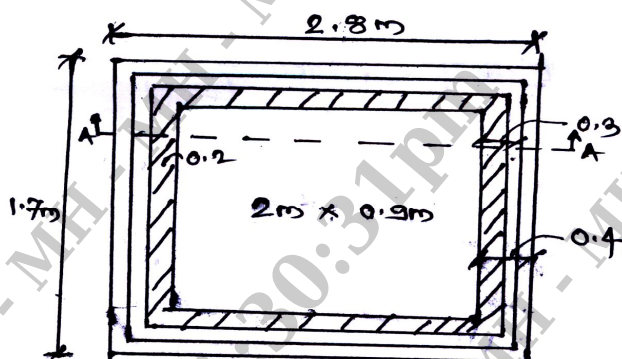
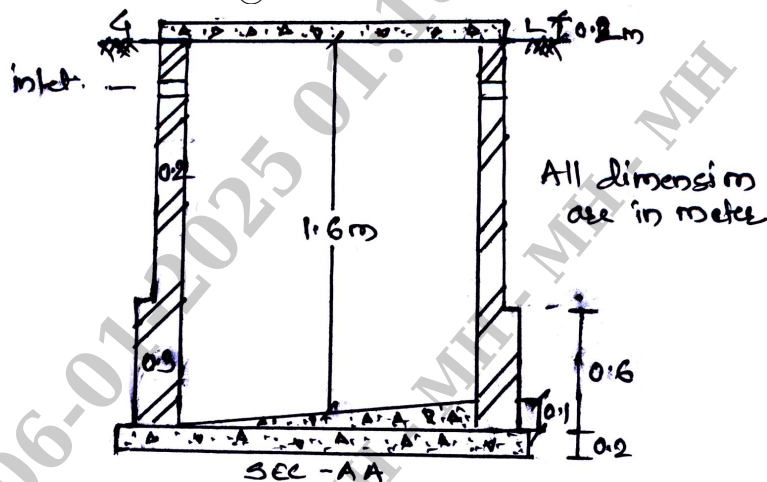
Fig.Q.2

(20 Marks)

Module-2

3 The details of a septic tank is as shown in Fig.Q.3. Estimate the quantities for the following items of work and work out the cost.

- Earth work excavation in foundation @ Rs.400/m³
- Cement concrete 1:3:6 floor and foundation @ Rs.4000/m³
- First class brick work with cement mortar 1:4 @ Rs.6000/m³.
- 12 mm thick plaster for side walls @ Rs.200/m².



PLAN.
Fig.Q.3

(20 Marks)

OR

4 Reduced level (RL of ground along centerline of proposed road from chainage 10 to 20 chainage are given below. The formation level at the 10th chainage is 107 and the road is in down word gradient of 1 in 150 upto the chainage 14 and the gradient changes to 1 in 100 down word gradient. Formation width is 10m and side slope of banking are 2:1 (H:V). Length of chain is 30 m. Estimate the quantities and cost of earth at the rate for filling is Rs.250 and cutting Rs.140 by mid sec. method.

Chainage	10	11	12	13	14	15	16	17	18	19	20
RL of the GL	105.0	105.60	105.44	105.90	105.42	104.30	105.00	104.10	104.62	104.00	103.30
RL of FL	107.00										
Gradient	Down word 1 in 150					Down word 1 in 100					

(20 Marks)

Module-3

- 5 a. Define specification. Explain the objectives of writing specification. (10 Marks)
 b. List the significances of microsoft excel in estimation. (10 Marks)

OR

- 6 Write detailed specification for following:
 i) Bed concrete for foundation with M7.5 CC
 ii) Size stone masonry for foundation in CM 1:6
 iii) Plastering 12 mm thick with CM 1:5
 iv) Floor filling with granite in CM 1:3 (20 Marks)

Module-4

- 7 a. Define rate analysis. Explain the purpose of rate analysis. (10 Marks)
 b. Work out the rate analysis for R.C.C. for slab and beam in CC 1:1.5:3 and steel 1.2%. (10 Marks)

OR

- 8 Analyze rates from first principle for following:
 i) Ashlar masonry for foundation in CM 1:6
 ii) BBM for super structure in CM 1:6
 iii) Plastering 20 mm thick in CM 1:4
 iv) Pointing in CM 1:3. (20 Marks)

Module-5

- 9 a. Define contracts. Discuss the essentials of contracts. (10 Marks)
 b. Write a note on tender documents and earnest money deposit. (10 Marks)

OR

- 10 Write short notes on following:
 a. Administrative approval and technical sanction
 b. Nominal muster roll
 c. Security deposit
 d. Running bills. (20 Marks)

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21CV72

Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Construction Technology for Substructures and Super Structures

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Write a note on Micro Tunnelling. (10 Marks)
b. Write a note on TBM. (10 Marks)

OR

- 2 a. What is bedding of conduits explain its significance. (10 Marks)
b. Explain different methods of compaction technology. (10 Marks)

Module-2

- 3 a. What is underwater drilling and blasting give one case study. (10 Marks)
b. Explain Jet grouting with a neat sketch. (10 Marks)

OR

- 4 a. What is the role of dewatering in shallow and deep excavation and list various method. (10 Marks)
b. Explain well point system with a neat sketch. (10 Marks)

Module-3

- 5 a. Write a note on operation and erection of R.M.C plant. (10 Marks)
b. Write a note on Concrete mix design. (10 Marks)

OR

- 6 a. Explain underwater concrete and its significance and also list the various methods used. (10 Marks)
b. Write a note on Self-Compacting Concrete. (10 Marks)

Module-4

- 7 a. Write a note on piles in land and marine structures. (10 Marks)
b. Explain initial and routine tests on piles as per codal specifications. (10 Marks)

OR

- 8 a. Write a note on pile driving by vibration and construction of micro piles. (10 Marks)
b. Write a note on Diaphragm walls. (10 Marks)

Module-5

- 9 a. What are cofferdams and explain the construction of a double wall cofferdam? (10 Marks)
b. State the uses of cofferdams and explain concrete wall movable cofferdams. (10 Marks)

OR

- 10 a. What are the differences between pneumatic and open caissons? (10 Marks)
b. Write a note on Well foundations. (10 Marks)

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Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Pavement Materials and Construction

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the desirable properties of road aggregates. (10 Marks)
- b. With a neat sketch, explain aggregate crushing test. (10 Marks)

OR

- 2 a. Explain the desirable properties of bitumen. (10 Marks)
- b. With a neat sketch, explain ductility test on bitumen. (10 Marks)

Module-2

- 3 a. Explain the desirable properties of bituminous mixes. (10 Marks)
- b. With the help of graph, explain Roth Futch's method for gradation of aggregates. (10 Marks)

OR

- 4 a. Explain in detail about Marshall mix design. (10 Marks)
- b. The specific gravities and weight proportions for aggregates and bitumen are as under for the preparation of Marshall mix design. The volume and weight of one Marshall specimen was found to be 475 cc and 1100 gms. Assuming absorption of bitumen in aggregates is zero. Find V_v , V_b , V_{MA} and V_{FB} .

Item	A-1	A-2	A-3	A-4	B
wt (gms)	825	1200	325	150	100
Specific gravity	2.63	2.51	2.46	2.43	1.05

(10 Marks)

Module-3

- 5 a. Write a short note on admixtures. (10 Marks)
- b. What are fibers in concrete? Explain different types of fibers in concrete. (10 Marks)

OR

- 6 a. Give explanation of fly ash usage in concrete. (10 Marks)
- b. Explanation on waste plastic in concrete. (10 Marks)

Module-4

- 7 a. What types of excavation equipment are used on construction site? With a neat sketch, explain power shovel its operation and application. (10 Marks)
- b. Explain the steps used in the preparation of subgrade. (10 Marks)

OR

- 8 a. Write an explanatory note on rollers in road construction. (10 Marks)
- b. List out the quality control tests for construction of subgrade. (10 Marks)

Module-5

- 9 a. Explain the construction steps for flexible pavements. (10 Marks)
b. Explanation on quality control tests on granular sub base. (10 Marks)

OR

- 10 a. Explain the construction steps for cement concrete pavements. (10 Marks)
b. What are the requirements of a good joint? Explain briefly with diagram of expansion joint in concrete. (10 Marks)

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21CV736

Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Repair, Retrofitting and Rehabilitation of Structures

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define and explain the following terms:
 - i) Repair
 - ii) Retrofitting
 - iii) Rehabilitation
 - iv) Strengthening
 - v) Deterioration.

(10 Marks)
- b. Explain the effect of earthquake on concrete structural elements, its evaluation for structural damages.

(10 Marks)

OR

- 2 a. Describe how acid attack on concrete cause deterioration of concrete structures.

(10 Marks)
- b. Explain briefly the durability of concrete and freezing and thawing.

(10 Marks)

Module-2

- 3 a. Explain the rapid structural assessment and investigation of damage, procedure performed on the concrete structures.

(10 Marks)
- b. Summarize the steps involved in damage assessment procedure with the help of flow chart for structure damage assessment.

(10 Marks)

OR

- 4 a. Discuss the testing systems of hardened concrete, by summarizing the various tests conducted on the concrete structural elements.

(10 Marks)
- b. Explain briefly the surface hardness and ultrasonic pulse velocity method of Non Destructive Testing system (NDTs).

(10 Marks)

Module-3

- 5 a. Describe the effects due to design and construction errors.

(12 Marks)
- b. Explain briefly the effects due to corrosion and nature of reinforcement corrosion mechanism.

(08 Marks)

OR

- 6 a. Explain the various corrosion protection techniques.

(12 Marks)
- b. Compare the effect of cover thickness and cracking on the serviceability and durability aspects of structures as per Indian standard code.

(08 Marks)

Module-4

- 7 a. Explain the types and necessities of maintenance of structures. (10 Marks)
b. Write a short note on:
i) Importance of Maintenance
ii) Need for Retrofitting. (10 Marks)

OR

- 8 a. Describe the Beam Jacketing Technique with a neat diagrams. (10 Marks)
b. Explain the process involved in near surface mounted technique. (10 Marks)

Module-5

- 9 a. Explain briefly types of fibers and advantages of fiber reinforced polymers. (08 Marks)
b. Summarize the advantages, disadvantages and applications of Sisal and Jute fibers. (12 Marks)

OR

- 10 a. Describe the types of special mortars. (08 Marks)
b. Explain the following briefly:
i) Guniting or Shotcreting
ii) Shoring
iii) Underpinning. (12 Marks)

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21CV742

Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Ground Improvement Technique

Time: 3 hrs.

Max. Marks: 100

Note : 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Assume missing data if any.

Module-1

- 1 a. Discuss about effect of compaction on Engineering properties of Soil. (10 Marks)
- b. Explain about construction of stone columns for deep compaction with neat sketch. (10 Marks)

OR

- 2 a. Explain how proctor needle is used to control compaction in field. (10 Marks)
- b. A proposed embankment fill required 12,000 m³ of compacted soil. The void ratio of compacted fill is specified as 0.65. The following table list 4 available borrow pits along with void ratio and cost per cum of moving the soil to proposed site. Assuming $G = 2.7$. Show the suitability of each site.

Borrow pit	Void ratio	Cost in Rs/m ³
A	0.9	60
B	1.1	50
C	0.95	80
D	0.75	10

(10 Marks)

Module-2

- 3 a. Explain about the Mechanism of soil – lime stabilization technique. (10 Marks)
- b. Discuss about Engineering behavior of soil – lime stabilization. (10 Marks)

OR

- 4 a. List the Engineering properties of fly ash. (10 Marks)
- b. Outline about change in properties of soil by calcium chloride stabilization. (10 Marks)

Module-3

- 5 a. Air mark the main object of Dewatering technique in soil. (10 Marks)
- b. Explain with neat sketch about Electro – Osmosis method of dewatering. (10 Marks)

OR

- 6 a. Write a note on seepage control in construction site. (10 Marks)
- b. List the various methods of dewatering. Explain any one method. (10 Marks)

Module-4

- 7 a. Discuss about the components of Reinforced Earth with typical sketch. (10 Marks)
- b. List the advantages of Reinforced Earth structure and also highlight about Application of soil reinforcement. (10 Marks)

OR

- 8 a. Explain step by step procedure of construction of soil nailing technique. (10 Marks)
b. List the advantages and disadvantages of soil nailing. (10 Marks)

Module-5

- 9 a. List the classification of Geotextiles. (10 Marks)
b. Explain about the applications of Geosynthetics. (10 Marks)

OR

- 10 a. Write a note on Anchor Rock bolts. (10 Marks)
b. Explain construction and advantages of Micropile with supporting sketch. (10 Marks)

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21CV743

Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Pavement Design

Time: 3 hrs.

Max. Marks: 100

**Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Use of IRC:37-2001 and IRC:58-2002 is permitted.**

Module-1

- 1 a. Explain the various components of pavements with their function with neat sketch. (08 Marks)
- b. Write the difference between flexible and rigid pavement. (06 Marks)
- c. Distinguish between Boussinesq's theory and Burmister theory. (06 Marks)

OR

- 2 a. Compare the salient features of flexible and rigid pavement. (06 Marks)
- b. Bring out the difference between highway and airport pavement. (06 Marks)
- c. Plate bearing tests were conducted using 30 cm diameter plates on soil subgrade and over a base course of thickness 45 cm. The pressure yielded at 0.5 cm deflection on the subgrade and base course were 1.25 kg/cm² and 8.0 kg/cm² respectively. Design the thickness requirement of flexible pavement for a wheel load at 5100 kg with tyre pressure of 7.0 kg/cm² for an allowable deflection of 0.5 cm using Burmister's two-layer deflection factor chart Fig.Q2(c).

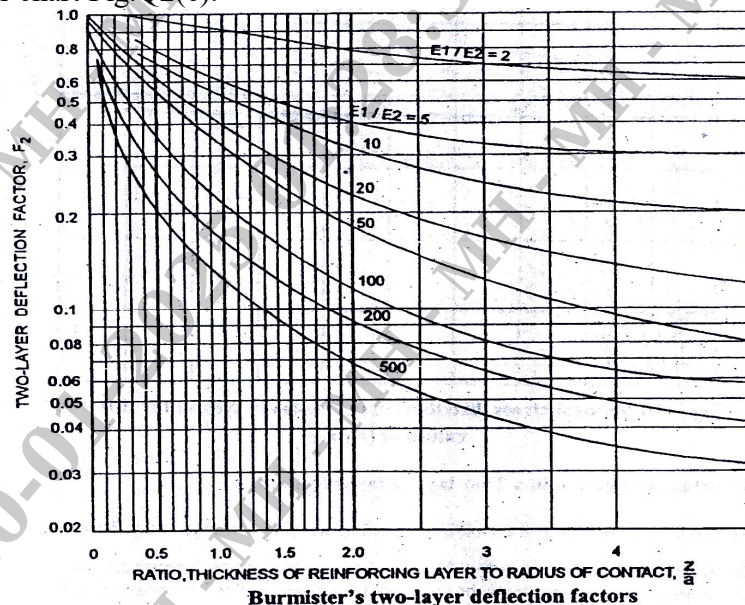


Fig.Q2(c)

(08 Marks)

Module-2

- 3 a. Explain the principle and design steps of McLeod method of pavement design. (10 Marks)
- b. Explain determination of ESWL by graphical method. Also draw the neat sketch. (10 Marks)

OR

- 4 a. The initial traffic after completion of construction of a four lane divided highway is estimated to be 3500 cv per day. Design the flexible pavement for a life of 15 years using the data given below:
Design CBR value = 8% , growth rate of cv = 6.5% , Average VDF value of cv = 4.0. Use IRC : 37 – 2001. (10 Marks)
- b. Explain the various factors to be considered for the design of pavement. (10 Marks)

Module-3

- 5 a. Write the principle of Benkelman beam test. Explain the procedure of determining the deflection value at any point on a flexible pavement. (08 Marks)
- b. Briefly explain the typical types of flexible pavement failures. (06 Marks)
- c. Explain functional evolution of pavement by unevenness index. (06 Marks)

OR

- 6 a. Write a short note on Falling Weight Deflectometer. (08 Marks)
- b. Explain briefly the various design factors for runway pavements. (06 Marks)
- c. Explain maintenance of Bitumenous surfaces. (06 Marks)

Module-4

- 7 a. As per IRC: 58-2002, explain the procedure of design of rigid pavements. (10 Marks)
- b. Determine the warping stresses at interior edge and corner of a 25cm thick cement concrete pavement with transverse joints at 5.0 interval and longitudinal joints at 3.6m intervals. The modulus of subgrade reaction, k is 6.9 kg/cm^3 and radius of loaded area is 15 cm. Assume maximum temperature differential during day to be 0.6°C per cm slab thickness and maximum temperature differential of 0.4°C per cm slab thickness during the night. Data are given below : $e = 10 \times 10^{-6}$ per $^\circ\text{C}$; $E = 3 \times 10^5 \text{ kg/cm}^2$; $\mu = 0.15$. (10 Marks)

OR

- 8 a. Explain design procedure for design of Tie Bars. (10 Marks)
- b. The design thickness of a CC pavement is 26 cm considering a design axle load (98th percentile load) of 12,000 kg on single axle and M-40 concrete with characteristic compressive strength of 400 kg/cm^2 . The radius of relative stiffness is found to be 62.2 cm. If the elastic modulus of dowel bar steel is $2 \times 10^6 \text{ kg/cm}^2$, modulus of dowel concrete interaction is 41500 kg/cm^3 and joint width is 1.8 cm. Design the dowel bars for 40% load transfer considering edge loading. (10 Marks)

Module-5

- 9 a. List and explain any five types of failures in rigid pavement. (10 Marks)
- b. With sketches, describe the various types of joints and their requirements, in rigid pavements. (10 Marks)

OR

- 10 a. Explain the desirable properties of subgrade. (10 Marks)
- b. Explain different methods of pavement evaluation. (10 Marks)

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Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Environmental Protection and Management

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Enumerate the various environmental management principles. Explain any two principles. (10 Marks)
- b. Explain abatement of pollution. List out the major activities initiated under the various scheme on pollution Abatement. (10 Marks)

OR

- 2 a. Write a note on Environmental stewardship. (10 Marks)
- b. Write a brief about the organizational drivers to the implementation of environmental management systems. (10 Marks)

Module-2

- 3 a. List and explain the environmental quality objectives. (10 Marks)
- b. Briefly explain :
 - i) Clean production and clean technology
 - ii) Concentration and mass standards
 (10 Marks)

OR

- 4 a. Explain the factors of CPCB. (10 Marks)
- b. Explain environmental performance indicators for an organization. (10 Marks)

Module-3

- 5 a. Enumerate the benefit and barriers in implementing ISO 14001 in an organization. (10 Marks)
- b. Write a note on :
 - i) Initial Environmental Review (IER)
 - ii) Management Review (MR)
 (10 Marks)

OR

- 6 a. Write a note on environmental aspects and impact with an organization setup. (10 Marks)
- b. Define environmental policy and write a note on developing environmental policy. (10 Marks)

Module-4

- 7 a. With an example, explain of environmental due-Diligence Audit. (10 Marks)
- b. Enumerate the contents of environmental statement (Form V) (10 Marks)

OR

- 8 a. Write a note on waste minimization planning in an industry. (10 Marks)
- b. Explain briefly :
i) Non conformity
ii) Roles and qualification of Auditors (10 Marks)

Module-5

- 9 a. Explain Transboundary movement of pollutant with an example. How to manage this problem. (10 Marks)
- b. Explain the pollution prevention methods adopted in an electroplating industry. (10 Marks)

OR

- 10 a. Discuss pollution prevention opportunities in pulp paper and sugar mills. (10 Marks)
- b. Briefly discuss the application of EMS and waste Audit. (10 Marks)

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Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Advanced Mathematical Methods

Time: 3 hrs.

Max. Marks: 100

Note : 1. Answer any FIVE full questions, choosing ONE full question from each module.
 2. Provide Normal distribution, t -distribution and F -distribution tables.

Module-1

- 1 a. Transform the matrix

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 1 & 2 \\ 4 & 2 & 1 \end{bmatrix}$$

into tri-diagonal form by Given's method. Using Strum's sequence ,

obtain exact eigen values or the interval of unit length each containing one eigen value.

(10 Marks)

- b. Perform two iterations of the Bridge – Vieta method to find a real root of the equation $x^4 - 3x^3 + 3x^2 - 3x + 2 = 0$. Use the initial approximation $P_0 = +0.5$ and also find the deflated polynomial. (10 Marks)

OR

- 2 a. Transform the matrix

$$A = \begin{bmatrix} 1 & 3 & 4 \\ 3 & 2 & -1 \\ 4 & 1 & 1 \end{bmatrix}$$

into tri – diagonal form by Householder's method. Using Strum's

sequence , obtain exact eigen value or the interval of unit length each containing one eigen value.

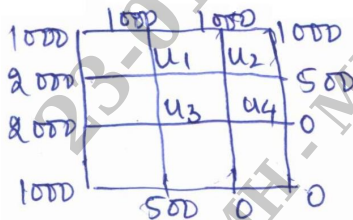
(10 Marks)

- b. Perform two iterations of the Bairstow method to extract a quadratic factor $x^2 + px + q$ from the polynomial $x^3 + x^2 - x + 2 = 0$. Use the initial approximations $P_0 = -0.9$, $q_0 = 0.9$.

(10 Marks)

Module-2

- 3 a. Solve the Laplace equation
- $U_{xx} + U_{yy} = 0$
- for the square mesh of the below figure with boundary values as given below :



Carryout three iterations.

(10 Marks)

- b. Solve the Poisson's equation $U_{xx} + U_{yy} = -81xy$. Where $0 < x < 3$, $0 < y < 3$ given that $u(0, y) = 0$, $u(x, 0) = 0$, $u(3, y) = 100$, $u(x, 3) = 100$ and $h = 1$.

(10 Marks)

OR

- 4 a. Use the Crank – Nicolson scheme to solve the heat equation problem

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} \text{ with the conditions : } u(x, 0) = \sin x, \quad 0 \leq x \leq \pi, \quad u(0, t) = u(\pi, t), \quad t > 0.$$

(10 Marks)

- b. Obtain the solution of $\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}$ by taking $h = 0.2$, where $0 \leq x \leq 1$, subjected to the initial conditions $u(x, 0) = \sin \pi x$, $u_t(x, 0) = \sin \pi x$ and the boundary conditions $u(0, t) = 0$, $u(1, t) = 0$ by using explicit method.

(10 Marks)

Module-3

- 5 a. Find the Taylor's expansion of $f(z) = \frac{2z^3 + 1}{z^2 + 1}$ in the region $z = i$. (06 Marks)

- b. Evaluate $\oint \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)^2(z-2)} dz$, where C is the circle $|z| = 3$. (07 Marks)

- c. Show that $\int_0^{2\pi} \frac{\cos 2\theta}{(1-2a \cos \theta + a^2)} d\theta = \frac{2\pi a^2}{(1-a^2)}$, $a^2 < 1$ by using contour integration. (07 Marks)

OR

- 6 a. Find the Laurent's expansion of $f(z) = \frac{7z-2}{(z-1)(z-2)}$ about the point $1 < z < 2$. (06 Marks)

- b. Evaluate $\oint_C \frac{z-3}{(z^2+2z+5)} dz$, where C is the circle $|z+1-i| = 2$. (07 Marks)

- c. Evaluate $\int_0^\infty \frac{\cos ax}{x^2+1} dx$ by using Contour integration. (07 Marks)

Module-4

- 7 a. Two independent sample of eight and seven items respectively had the following values of the variable :

Sample 1 :	9	11	13	11	15	9	12	14
Sample 2 :	10	12	10	14	9	8	10	

Do the two estimates of population variance differ significantly at 5% level of significance ($F_{0.05} = 4.21$). (10 Marks)

- b. Set up an analysis of variance (ANOVA) table to assess the significance of possible variation in performance in a certain test between the schools of a city. A common test was given to a numbers of students taken at random from the fifth class of the three schools concerned the results given below :

A	B	C
9	13	14
11	12	13
13	10	17
9	15	7
8	5	9

Construct the analysis of variance for the given data. (10 Marks)

OR

- 8 a. The mean and S.D of the maximum loads supported by 60 cables are 11.09 tonnes and 0.73 tonnes respectively. Find i) 95% ii) 99% confidence limits for mean of the maximum loads of all cables produced by the company. (10 Marks)
- b. Measurements on the length of a copper were taken by two experiments A & B are as under

A's measurements	12.29	12.25	11.86	12.13	12.44	12.78	12.77	11.90	12.47
B's measurements	12.39	12.46	12.34	12.22	11.98	12.46	12.23	12.06	

Test whether B's measurements are more accurate than A's at 5% level of significance.

(10 Marks)

Module-5

- 9 a. Explain i) Absorbing state ii) Transient state iii) Recurrence state. (06 Marks)
- b. Three boys A, B and C are throwing a ball to each other. A always throws the ball to B and B always throws the ball to C. But C is just as likely to throw the ball to B as to A. If C was the first person to throw the ball, find the probability that A has the ball for the forth throw. (07 Marks)
- c. Customers arrive at a one – man barber shop according to a Poisson process with a mean inter arrival time of 12 min. Customers spend an average of 10mm in the barber's chair
- i) What is the expected number of customers in the barber shop and in queue?
- ii) Calculate the percentage of time an arrival can walk straight into barber's chair without having to wait. (07 Marks)

OR

- 10 a. Find the unique fixed probability vector for the regular stochastic matrix.

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

(06 Marks)

- b. A student's study habits are as follows. If he studies one night , he is 70% sure not to study the next night. On the other hand , if he does not study one night , he is 60% sure not to study the next night. In the long run how often does he study? (07 Marks)
- c. Given an average arrival rate of 20 per hour, is it better for a customer to get service at a single channel with mean service rate of 22 customers per hour or at one of two channels is parallel with mean service rat at 11 customers per hour for each of the two channels. Assume both the queues to be Poisson type. (07 Marks)

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CBCS SCHEME

USN

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18CV32

Third Semester B.E. Degree Examination, Dec.2024/Jan.2025 Strength of Materials

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain in brief St. Venant's principle. (04 Marks)
- b. Derive an expression for the deformation of a circular bar tapering uniformly subjected to axial tensile force 'P'. (08 Marks)
- c. Find the elongation of the bar shown in Fig.Q1(c). Take $E = 210 \text{ GN/m}^2$.

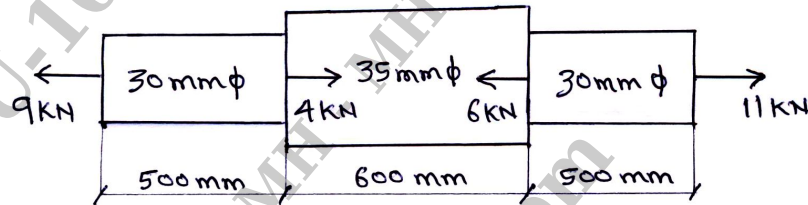


Fig.Q1(c)

(08 Marks)

OR

- 2 a. Write a short note on temperature stress in simple bars. (05 Marks)
- b. A reinforced concrete column of size $230 \text{ mm} \times 400 \text{ mm}$ has 8 steel bars of 12 mm diameter distributed evenly on all sides. If the column is subjected to an axial compression of 600 kN, find the stresses developed in steel and concrete. Modular ratio = 18.67. (07 Marks)
- c. A compound bar is made up of a central steel plate 60 mm wide and 10 mm thick to which copper plates 40 mm wide by 5 mm thick are connected rigidly on each side. The length of the bar at normal temperature is 1 metre. If the temperature is raised by 80°C , determine the stresses in each metal and change in length.

Take $E_s = 200 \text{ GN/m}^2$ $\alpha_s = 12 \times 10^{-6}/^\circ\text{C}$

$E_c = 100 \text{ GN/m}^2$ $\alpha_c = 17 \times 10^{-6}/^\circ\text{C}$

(08 Marks)

Module-2

- 3 a. Discuss briefly compound stresses in beams. (06 Marks)
- b. Explain the procedure for determining stresses in a general two dimensional stress system using Mohr's circle. (06 Marks)
- c. The principal stresses at a certain point in a strained material are 150 N/mm^2 and 48 N/mm^2 both tensile. Find the normal and tangential stresses on a plane inclined at 20° with the major principal plane. Also determine the resultant stress, obliquity and maximum shear stress. (08 Marks)

OR

- 4 a. In a thin cylinder show that the hoop stress is twice the longitudinal stresses. (08 Marks)
- b. A thick cylinder of external and internal diameter of 300 mm and 180 mm is subjected to an internal pressure of 42 N/mm^2 and external pressure 6 N/mm^2 . Determine the stresses in the material. Now if the external pressure is doubled, what internal pressure can be maintained without exceeding the previously determined maximum stress? (12 Marks)

Module-3

- 5 a. Establish the relationship between load intensity, SF and BM. (04 Marks)
- b. Draw SFD and BMD for the beam shown in Fig.Q5(b).

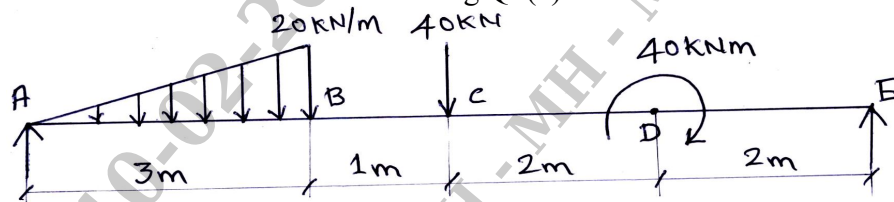


Fig.Q5(b)

(08 Marks)

- c. Draw SFD and BMD for the beam shown in Fig.Q5(c).

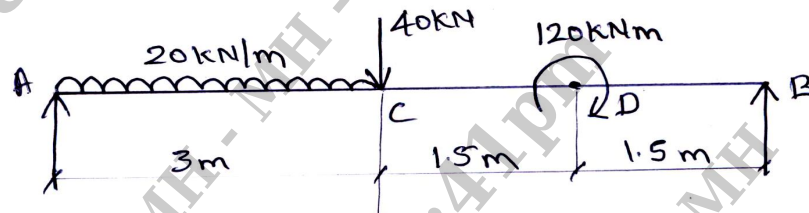


Fig.Q5(c)

(08 Marks)

OR

- 6 a. Explain in brief shear force and bending moment. (04 Marks)
- b. Draw SFD and BMD for the beam shown in Fig.Q6(b). Determine the position at which the positive BM occurs. Find also magnitude of the maximum positive and negative bending moment.

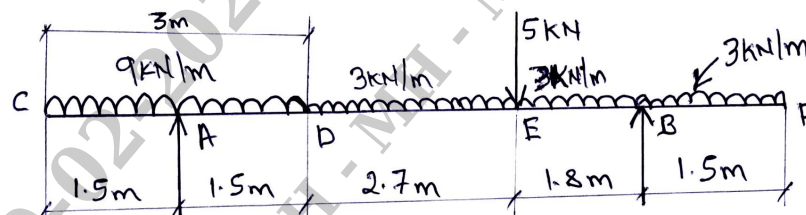


Fig.Q6(b)

(10 Marks)

- c. Draw SFD and BMD for the beam shown in Fig.Q6(c).

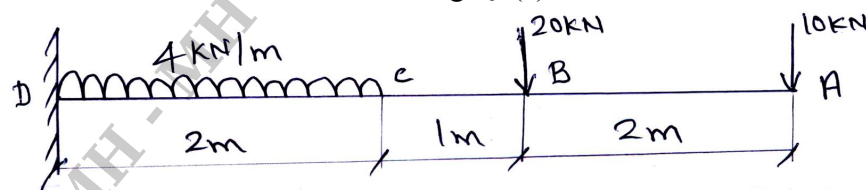


Fig.Q6(c)

(06 Marks)

Module-4

- 7 a. Explain in brief the theory of simple bending. (04 Marks)
- b. Derive the equation of pure bending with usual notations. (10 Marks)
- c. A cantilever beam of span 1 m has rectangular c/s of size 200 mm × 400 mm. Determine the concentrated load which placed at the free end produce shear stress intensity of 1.5 N/mm². Hence compute maximum bending stresses in the c/s at the fixed end of cantilever. (06 Marks)

OR

- 8 a. Prove that a hollow is stronger and stiffer than the solid shaft of same material, length and weight. (10 Marks)
- b. A shaft is required to transmit 245 KW @ 240 RPM. The maximum torque may be 1.5 times the mean torque. The shear stress in the shaft should not exceed 40 N/mm² and the twist 1° per metre length. Determine the diameter required if (i) Shaft is solid. (ii) Shaft is hollow with external diameter twice the internal diameter. (10 Marks)

Module-5

- 9 a. Determine the expression for slope and deflection in a simply supported beam subjected to UVL throughout, with loading intensity maximum at the left support and zero at the right support. (08 Marks)
- b. A cantilever beam AB 2 m long is carrying a load of 20 kN at free end and 30 kN at a distance of 1 m from the free end. Find slope and deflection at the free end. Take $E = 200 \text{ GPa}$ and $I = 150 \times 10^6 \text{ mm}^4$. (08 Marks)
- c. Define (i) Buckling load (ii) Slenderness ratio (04 Marks)

OR

- 10 a. Write the assumptions made in Euler's theory. (04 Marks)
- b. Distinguish between long and short column. (04 Marks)
- c. Determine the section of a cast iron hollow cylindrical column 3 m long with both ends firmly built in, if it carries an axial load of 800 kN. The ratio of internal to external diameter is 5/8. Use a factor of safety 4. Take $f_c = 550 \text{ N/mm}^2$ and Rankine's constant for both ends hinged case = 1/1600. (12 Marks)

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CBCS SCHEME

USN

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18CV33

Third Semester B.E. Degree Examination, Dec.2024/Jan.2025

Fluid Mechanics

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define the following fluid, properties and mention their SI units :
 i) Specific weight
 ii) Kinematic viscosity
 iii) Surface extension. (06 Marks)
- b. Explain the phenomenon of capillarity. Derive an expression for capillary rise of a liquid. (06 Marks)
- c. A differential manometer is connected at two points A and B as shown in Fig.Q1(c). At B, air, pressure is 9.81 N/cm^2 . Find the pressure at A. (08 Marks)

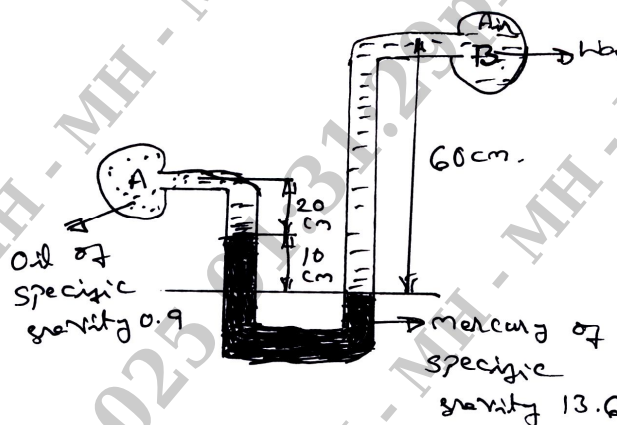


Fig.Q1(c)

OR

- 2 a. State and prove Pascal's law. (06 Marks)
- b. Clearly distinguish between :
 i) Absolute pressure
 ii) Gauge pressure
 iii) Vacuum pressure
 iv) Atmospheric pressure.
 Indicate their relative positions on a chart. (06 Marks)
- c. The dynamic viscosity of oil used for lubrication between a shaft and sleeve is 6 poise. The shaft is of diameter 0.4 m and it rotates at 190 rpm. Calculate the power lost in the bearing for a sleeve length of 90 mm. The thickness of the oil film is 1.5 mm. (08 Marks)

Module-2

- 3 a. Define :
 i) Total pressure
 ii) Center of pressure. (04 Marks)
- b. Derive an expression for total pressure and centre of pressure for a vertical plane surface submerged in a liquid. (08 Marks)
- c. A 30 cm diameter pipe conveying water branches into two pipes of diameter 20 cm and 15 cm respectively. If the average velocity in the 30 cm diameter pipe is 2.5 m/s, find the discharge in this pipe. Also determine the velocity in 15 cm pipe. If the average velocity in 20 cm diameter pipe is 2 m/s. (08 Marks)

OR

- 4 a. Define the following types of flow :
 i) Steady and unsteady flow
 ii) Laminar and turbulent flow
 iii) Compressible and incompressible flow. (06 Marks)
- b. Derive the continuity equation for a three dimensional steady, incompressible flow in Cartesian coordinates. (08 Marks)
- c. Determine the total pressure and centre of pressure of an isosceles triangular plate of base 4 m, altitude 4 m when it is immersed vertically in an oil of specific gravity 0.9. The base of the plate coincides with the free surface of oil. (06 Marks)

Module-3

- 5 a. Derive Euler's equation of motion along a stream line. Obtain Bernoulli's equation from Euler's equation. Mention the assumptions made. (08 Marks)
- b. Explain the working principle of Pitot tube with a sketch. (04 Marks)
- c. A horizontal venturimeter with inlet diameter 20 cm and throat diameter 10 cm is used to measure the flow of water. the pressure at inlet is 17.658 N/cm^2 and the vacuum pressure at the throat is 30 cm of Hg. Find the discharge of water. take $C_d = 0.98$. (08 Marks)

OR

- 6 a. State and explain the principle of momentum equation. (04 Marks)
- b. Derive an expression for discharge through a venturimeter. (08 Marks)
- c. A pipe line carrying oil of specific gravity 0.87 changes in diameter from 200 mm at position A to 500 mm at a position B which is 4 m at higher level. If the pressure at A and B are 10 N/cm^2 and 6 N/cm^2 respectively and discharge is 200 liters/s, determine the loss of head and indicate the direction of flow. (08 Marks)

Module-4

- 7 a. Derive an expression for discharge over a triangular notch. (08 Marks)
- b. Write a note on classification of mouthpieces. (04 Marks)
- c. Determine the height of a rectangular weir of length 6 m to be built across a rectangular channel if the maximum depth of water on the upstream side of weir is 1.8 m and discharge is 2000 litres/s. Take $C_d = 0.6$ and neglect end contractions. (08 Marks)

OR

- 8 a. Define the hydraulic coefficients C , C_d and C_v of an orifice and obtain the relation between them. (08 Marks)
- b. Explain ventilation of weirs. (04 Marks)
- c. The head of water over an orifice of diameter 100 mm is 10 m. The water coming out from the orifice is collected in a circular tank of diameter 1.5 m. The rise of water level in the tank is 1 m in 25 seconds. The coordinates of a point of the jet measured from vena contracta are 4.3 m horizontal and 0.5 m vertical. Find the coefficients C_d , C_c and C_v . (08 Marks)

Module-5

- 9 a. Derive Darcy – Weisbach equation for head loss due to friction in a pipe. (08 Marks)
- b. Mention major and minor losses in a pipe flow. (04 Marks)
- c. Water is required to be supplied to a colony of 4000 residents at a rate of 180 liters per person from a source 3 km away. If half the daily requirement needs to be pumped in 8 hours against a frictional head of 18 m, find the size of the main pipe supplying water. Assume coefficient of friction as 0.007. (08 Marks)

OR

- 10 a. Explain the reasons for water hammering in pipes. (06 Marks)
- b. Obtain an expression for the rise of pressure when the flowing water in a pipe is brought to rest by sudden closure of valve and the pipe is elastic. (08 Marks)
- c. Find the loss of head when a pipe of diameter 200 mm is suddenly enlarged to a diameter of 400 mm. The rate of flow of water through the pipe is 250 liters/s. (06 Marks)

Fifth Semester B.E. Degree Examination, Dec.2024/Jan.2025

Analysis of Indeterminate Structures

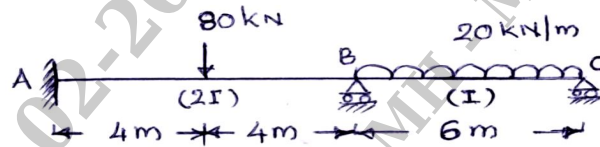
Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

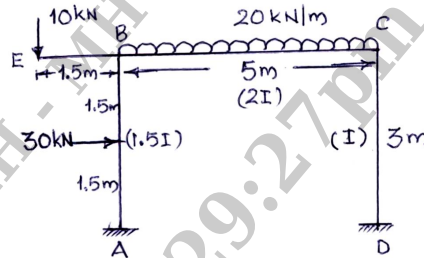
Module-1

- 1 Analyze the continuous beam shown in Fig.Q1 by slope deflection method. Draw BMD, SFD and elastic curve. (20 Marks)



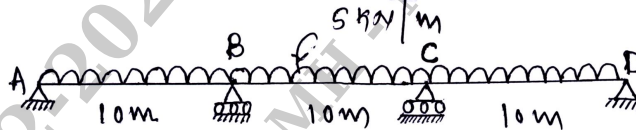
OR

- 2 Analyze the portal frame shown in Fig.Q2 by slope deflection method. Draw BMD and elastic curve. (20 Marks)



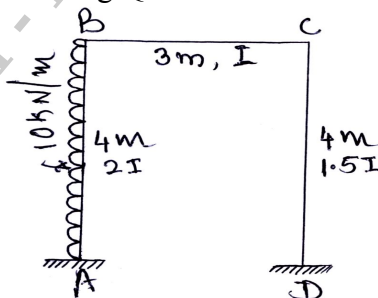
Module-2

- 3 Analyze the continuous beam ABCD loaded as shown in Fig.Q3 if settlement in support B and C are 5 mm and 10 mm respectively. Use moment distribution method. Take $EI = 2.7 \times 10^5 \text{ kN-m}^2$. Draw BMD. (20 Marks)



OR

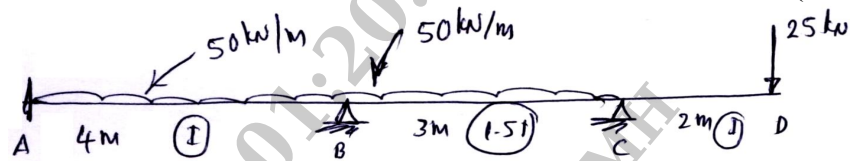
- 4 Analyze the frame loaded as shown in Fig.Q4. Use moment distribution method. (20 Marks)



Module-3

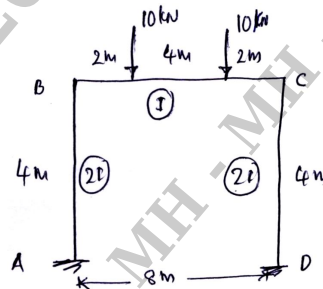
- 5 Analyze the continuous beam loaded shown in Fig.Q.5 by Kani's rotation method. Draw BMD. (20 Marks)

Fig. Q5

**OR**

- 6 Analyze the frame shown in Fig.Q.6 by Kani's method. Take the advantage of symmetry. (20 Marks)

Fig. Q6

**Module-4**

- 7 Analyze the continuous beam by using flexibility matrix method. Draw BMD, SFD and elastic curve. Refer Fig.Q7. (20 Marks)

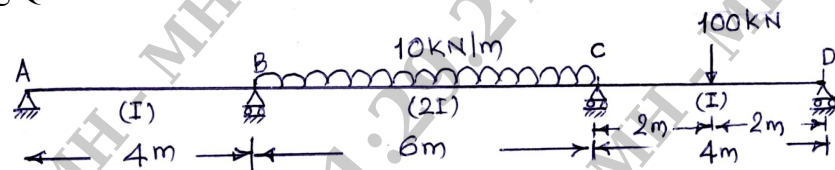


Fig. Q7

OR

- 8 Analyze the truss shown in Fig.Q8 by flexibility matrix method choosing force in the member AD as redundant. Assume constant EI for all the members. (20 Marks)

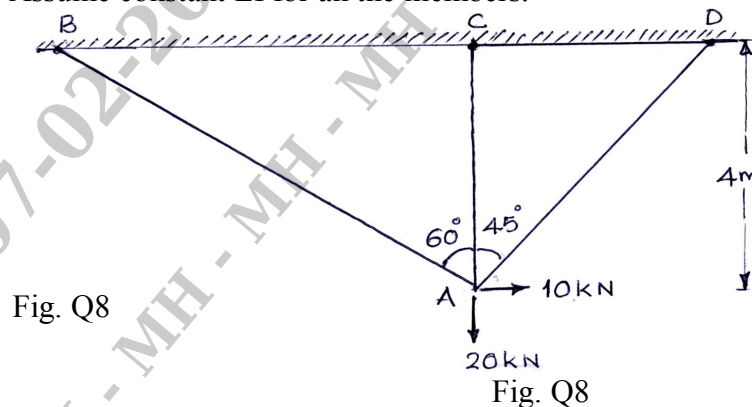


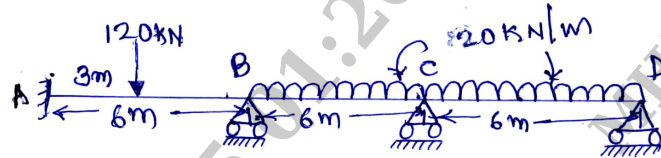
Fig. Q8

Fig. Q8

Module-5

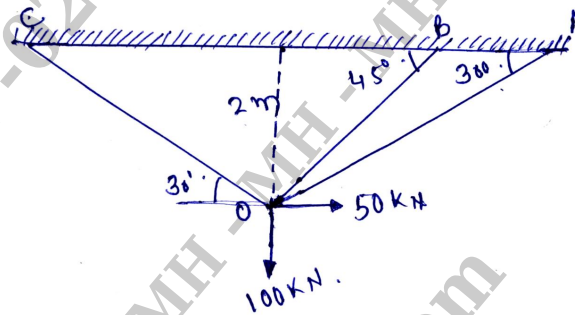
- 9 Analyze the continuous beam shown in Fig.Q9 by stiffness matrix method. Take EI constant. (20 Marks)

Fig. Q9

**OR**

- 10 Analyze the pin-jointed truss shown in Fig.Q10 by stiffness matrix method. Take cross-sectional area for all members = 1000 mm^2 and $E = 200 \text{ kN/mm}^2$. (20 Marks)

Fig. Q10



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Fifth Semester B.E. Degree Examination, Dec.2024/Jan.2025

Design of RC Structural Elements

Time: 3 hrs.

Max. Marks: 100

**Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Use of IS:456-2000, SP-16 is permitted.**

Module-1

- 1 a. Briefly explain the philosophy and principles of limit state method of RCC design. (08 Marks)
- b. Write a brief notes on:
 - i) Characteristic loads
 - ii) Characteristic strength
 - iii) Balanced section
 - iv) Over reinforced section. (12 Marks)

OR

- 2 a. List the factors that affect the deflection of reinforced concrete members. (04 Marks)
- b. A rectangular section beam 200 mm × 450 mm is reinforced with 3 bars of 16 mm diameter at an effective depth of 420 mm and with 2 bars of 12 mm hanger bars. The effective span of the beam is 5 m. The beam supports a service live load of 10 kN/m. If $f_{ck} = 20 \text{ N/mm}^2$, $f_y = 415 \text{ N/mm}^2$. Compute: i) The short term deflection ii) The long term deflection as per IS:456 code specifications. (16 Marks)

Module-2

- 3 a. A doubly reinforced concrete beam having a rectangular section 250 mm × 540 mm overall depth is reinforced with 2 bars of 12 mm hanger bars and 4 bars of 20 mm diameter in tension zone. Use effective cover as 40 mm, M20 grade concrete, Fe 415 HYSD bars. Estimate the flexural strength of section as per codal provisions. (10 Marks)
- b. A reinforced concrete beam of rectangular section has a width of 250 mm and an effective depth of 500 mm. The beam is reinforced with 4 bars of 25 mm diameter on the tension side. Two of the tension bars are bent up @ 45° near the support section. In addition the beam is provided with two legged stirrups of 8 mm diameter at 150 mm c/c. If $f_{ck} = 25 \text{ N/mm}^2$, $f_y = 415 \text{ N/mm}^2$. Calculate ultimate shear strength of the section. (10 Marks)

OR

- 4 a. Determine the area of tensile reinforcement required in a flanges beam having width of flange 750 mm, width of rib 300 mm, thickness of flange 120 mm, effective depth 600 mm to support a factored moment of 300 kN-m. Use M-20 grade concrete and Fe-415 HYSD bars. (10 Marks)
- b. A reinforced concrete beam of rectangular section 300 mm × 600 mm is reinforced with 4 bars of 25 mm diameter at an effective depth of 550 mm. The effective span of the beam is 7 m, take $f_y = 415 \text{ N/mm}^2$, $f_{ck} = 20 \text{ N/mm}^2$. Find the uniformly distributed ultimate load on the beam. (10 Marks)

Module-3

- 5 Design a singly reinforced concrete beam of clear span 5 m to support a design working live load of 10 kN/m. Adopt M-20 grade concrete and Fe-415 HYSD bars. Sketch reinforcement details. (20 Marks)

OR

- 6 A T-beam slab floor of reinforced concrete has a slab 150 mm thick spanning between the T-beams which are spaced 3 m c/c. The beams have a clear span of 10 m and the end bearings are 450 mm thick walls. The live load acting on the floor is 4 kN/m². Using M-20 grade concrete, Fe 415 HYSD bars, design one of the intermediate T-beams. (20 Marks)

Module-4

- 7 Design a simply supported R.C.C slab for an office floor having clear dimensions of 4 m × 10 m with 230 mm wall all-round. Adopt M-20 grade concrete and Fe-415 HYSD bars. Show reinforcement details. (20 Marks)

OR

- 8 a. Distinguish between one way slab and two way slab. (08 Marks)
b. Design a dog-legged flight of stairs of a commercial building spanning between landing beams. The flight has 12 No.s steps with tread 300 mm, rise 160 mm, width of landing beams 400 mm. Use M20 grade concrete and Fe 415 HYSD bars. Sketch the reinforcement details. (12 Marks)

Module-5

- 9 a. List out different types of columns. (04 Marks)
b. Design the reinforcements in a circular column of diameter 300 mm with helical reinforcement to support a factored load of 1500 kN. The column has an unsupported length of 3 m. Use M-20 grade concrete and Fe-415 HYSD bars. (16 Marks)

OR

- 10 Design a reinforced concrete footing for a rectangular column of section 300 mm × 500 mm supporting an axial factored load of 1500 kN. The SBC of the soil @ site is 185 kN/m². Use M-20 grade concrete and Fe 415 HYSD bars. (20 Marks)

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Fifth Semester B.E. Degree Examination, Dec.2024/Jan.2025 Basic Geotechnical Engineering

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Missing data may be assumed suitably.

Module-1

- 1 a. Starting from three phase diagram, with usual notations deduce.
- $$\gamma_b = \frac{(G + S_r e) \gamma_w}{(1 + e)}$$
- and hence from this equation obtain the expressions for γ_d , γ_{sat} and γ_{sub} . (10 Marks)
- b. A saturated specimen of undisturbed inorganic clay has a volume of 19.20 cm³ and mass 32.50 gm. After overdrying at 105°C for 24 hrs, the mass reduces to 20.90 gm. For the soil in the natural state find the following:
- Water content
 - Dry unit weight (γ_d)
 - Specific gravity of soil solids
 - Voids ratio
 - Saturated unit weight (γ_{sat}). (10 Marks)

OR

- 2 a. With a neat sketch, explain Atterberg limits. (06 Marks)
- b. Explain with a neat sketch the particle size distribution curve. (04 Marks)
- c. A soil sample consisting of particles of size ranging from 0.5 mm to 0.01 mm is put on the surface of still water tank of 5 m deep. Calculate the time of settlement of the coarsest and finest particles of the sample, to the bottom of the tank. Assume average specific gravity of soil particles as 2.66 and viscosity of water as 0.01 poise. (10 Marks)

Module-2

- 3 a. Explain with sketches the following clay minerals:
- Kaolinite
 - Illite
 - Montmorillonite
- b. With sketches explain briefly primary and secondary bonds. (05 Marks)
- c. With a neat sketch, explain diffuse double layer in fine grained soils. (06 Marks)

OR

- 4 a. Discuss compaction of soils. Distinguish between standard proctor and modified proctor compaction tests. (06 Marks)
- b. With a neat sketch, explain zero air voids line. (04 Marks)
- c. The following results refer to a standard compaction test:

Water content (%)	10	12	14.3	16.1	18.2
Weight of soil with mould (N)	29.25	30.95	31.50	31.25	30.70

Determine the optimum moisture context and maximum dry unit weight by plotting the compaction curve. Take $G = 2.70$, volume of mould = 1000 cm³ and weight of mould = 10N. (10 Marks)

Module-3

- 5 a. Explain the following:
 i) Superficial velocity
 ii) Seepage velocity. (04 Marks)
- b. Explain briefly factors affecting permeability. (06 Marks)
- c. In a falling head permeability test, the soil sample used is 20 cm long with a cross sectional area 24 cm^2 . Calculate the time required for the head causing flow to drop from 250 mm to 120 mm. The cross sectional area of the stand pipe is 2 cm^2 . The soil sample is made up of 3 layers.
 The thickness of first layer from the top is 8 cm and has a value of K as $2 \times 10^{-4} \text{ cm/sec}$. The second layer has thickness of 7 cm and it has $K = 5 \times 10^{-4} \text{ cm/sec}$. The bottom most layer has a K value of $7 \times 10^{-4} \text{ cm/sec}$. Flow is in a direction perpendicular to the layers. (10 Marks)

OR

- 6 a. What is a flow net? What are the uses and characteristics of flow nets? (06 Marks)
- b. Calculate the seepage loss in m^3/day for a hydraulic structure, if the flow net contains 5 flow lines and 9 equipotential lines and the head causing flow is 20 m. K of soil is $2.6 \times 10^{-6} \text{ cm/sec}$. (04 Marks)
- c. The water table in a deposit of sand 10 m thick, is at a depth of 4 m below the surface. Above the water table, the sand is saturated with capillary water. The bulk unit weight of sand is 19 kN/m^3 . Calculate the effective pressures at 2 m, 4 m and 10 m below the surface. Hence plot the variation of total pressure, neutral pressure and effective pressure over the depth of 10 m. (10 Marks)

Module-4

- 7 a. Explain Mohr-Coulomb's failure theory and draw the failure envelope for different soils. (08 Marks)
- b. In a triaxial test on two identical soil samples, the following data was obtained.

Test No	Cell pressure (kN/m^2)	Maximum deviator stress (kN/m^2)	Maximum principal stress (kN/m^2)
1	50	120	-
2	100	-	332

Compute shear parameters.

(12 Marks)

OR

- 8 a. Mention the advantages and disadvantages of direct shear test. (05 Marks)
- b. Classify shear tests based on drainage conditions. (03 Marks)
- c. A soil has unconfined compression strength of 120 kN/m^2 . In triaxial compression test, specimen of same soil (under similar conditions) when subjected to cell pressure of 40 kN/m^2 , failed at an additional stress of 160 kN/m^2 . Determine:
 i) Shear strength parameters
 ii) Angle made by failure plane with axial stress direction in case of triaxial test. (12 Marks)

Module-5

- 9 a. Explain with neat sketches the mass spring analogy of consolidation of soils. (08 Marks)
- b. Explain normally consolidated soil and over consolidated soil. (06 Marks)
- c. The thickness of a normally consolidated clay layer is 3.0 m. The initial voids ratio of the sample is 1.0 and its liquid limit is 60%. The effective over burden pressure at the middle of the clay layer was 154 kN/m^2 . Due to construction of a building the increase in effective stress is 92.4 kN/m^2 . Determine the consolidation settlement of the clay layer. (06 Marks)

OR

- 10 a. What are the assumptions made in Terzaghi's theory of one-dimensional consolidation? (04 Marks)
- b. Explain the significance of pre-consolidation pressure. Describe the Casagrande's method of determining it. (08 Marks)
- c. In a consolidation test voids ratio decreased from 0.70 to 0.60, when the pressure changes from 50 kN/m² to 100 kN/m². Determine:
- i) Compression Index
 - ii) Coefficient of compressibility
 - iii) Coefficient of volume change. (08 Marks)

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18CV55

Fifth Semester B.E. Degree Examination, Dec.2024/Jan.2025 Municipal Waste Water Engineering

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Missing data, if any, may be suitably assumed.

Module-1

- 1 a. Briefly explain the different types of water carriage system with their merits and demerits. (08 Marks)
- b. Define dry weather flow and explain the factors affecting Dry Weather Flow (DWF). (06 Marks)
- c. A certain district of a city has a projected population of 50,000 residing over an area of 40 hectares. Find the design discharge for the sewer line for the following data :
Rate of water supply = 200 Lpcd
Average impermeability coefficient = 0.3
Time of concentration = 30 minutes
The sewer line is to be designed for a flow equivalent to wet weather flow (WWF) plus twice the DNF. Use U.S ministry of health formula. Assume that 75% of water supply reaches in sewer as wastewater. (06 Marks)

OR

- 2 a. With a neat sketch explain construction and working of manhole. (08 Marks)
- b. Briefly explain the version tests conducted on sewer line. (06 Marks)
- c. Briefly explain the basic principles of home drainage system. (06 Marks)

Module-2

- 3 a. Briefly explain self cleansing velocity and non-scouring velocity along with their values. (06 Marks)
- b. Explain types of sampling and in which situations the types of sampling are used. (06 Marks)
- c. A stoneware sewer 30 cm in diameter is laid at gradient of 1 in 100 using $N = 0.013$ in Manning's formula, calculate the velocity discharge and Chezy's coefficient when the sewer is running full. (08 Marks)

OR

- 4 a. With the help of general flow diagram of municipal wastewater treatment, explain the function of various units. (08 Marks)
- b. Briefly explain the physical, chemical and biological characteristics of wastewater. (06 Marks)
- c. Define BOD and derive an expression for first stage BOD formation. (06 Marks)

Module-3

- 5 a. Define screening and explain types of screens. (06 Marks)
 b. With the help of neat sketch, explain the working of circular shape of settling tank. (08 Marks)
 c. Define a primary settling tank of rectangular shape for a town having a population of 50,000 with a water supply of 180 lit per capita per day. (06 Marks)

OR

- 6 a. Briefly explain the factors affecting self purification process. (08 Marks)
 b. Explain with a neat sketch, the salient features of oxygen sag curve. (06 Marks)
 c. What is meant by sewage sickness and explain the preventive measures adopted to avoid the same. (06 Marks)

Module-4

- 7 a. Briefly explain suspended growth system and attached growth system of wastewater treatment with example. (06 Marks)
 b. With the help of flow diagram, explain Activated Sludge Process (ASP) for treatment of wastewater. (06 Marks)
 c. Mention the modification of ASP and explain any two of them. (08 Marks)

OR

- 8 a. With a neat sketch explain the working of trickling filter. (08 Marks)
 b. Write short notes on :
 i) Mechanism of anaerobic sludge digestion
 ii) Oxidation ponds (06 Marks)
 c. With the help of neat sketch, explain the working of sludge drying beds. (06 Marks)

Module-5

- 9 a. Discuss in brief biological nitrification denitrification process for the removal of nitrogen from wastewater. (08 Marks)
 b. Explain removal of phosphorus from chemical precipitation method with equation. (06 Marks)
 c. Explain electro coagulation method for treatment of wastewater. (06 Marks)

OR

- 10 a. With a neat sketch explain the working of septic tank. (08 Marks)
 b. Write short notes on :
 i) Eco – toilet
 ii) Soak pits. (06 Marks)
 c. Design the dimension of a septic tank for a small colony of 150 persons provided with an assured water supply of 120 Lpcd. Assume the required data suitably. (06 Marks)

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Seventh Semester B.E. Degree Examination, Dec.2024/Jan.2025

Air Pollution and Control

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define Air pollution. Explain primary and secondary air pollutants with examples. (08 Marks)
- b. Explain the classification of Air pollution. (08 Marks)
- c. Explain various types of inversion. (04 Marks)

OR

- 2 a. State the effect of air pollution on health and vegetation. (10 Marks)
- b. Briefly explain photo chemical smog and with its effects. (10 Marks)

Module-2

- 3 a. Explain :
i) Temperature lapse rate ii) stability iii) wind velocity. (09 Marks)
- b. Explain the plume behavior with neat diagram. (11 Marks)

OR

- 4 a. Explain Wind Rose diagram. (06 Marks)
- b. State Meteorological variables. (06 Marks)
- c. A factory uses 2,00,000 litres of furnace oil (specific density 0.97) per month. If for one million litres of oil used per year, the particulate matter emitted is 3.0 tonnes per year, SO₂ emitted is 59.7 tonnes per year, NO_x emitted is 7.5 tonnes per year, hydrocarbons emitted are 0.37 tonnes per year and carbon monoxide emitted is 0.52 tonnes per year, calculate the height of the chimney required to be provided for safe dispersion of the pollutants. Calculate as per central Board for prevention and control of water pollution (1984), formula. (08 Marks)

Module-3

- 5 a. Explain the preliminary consideration and stages of sampling. (10 Marks)
- b. Explain the devices for sampling gaseous pollutants. (10 Marks)

OR

- 6 a. Explain Monitoring of Air pollutants. (10 Marks)
- b. Explain the Air quality models – Gaussian model briefly. (10 Marks)

Module-4

- 7 a. Explain with a neat diagram setting chamber with merits and demerits. (10 Marks)
- b. Explain Filters and ESP. (10 Marks)

OR

- 8 a. Explain the cyclone separations with a neat diagram, also with merits and demerits. (10 Marks)
- b. Explain the site selection for Industrial plant location. (10 Marks)

Module-5

- 9 a. Explain Air pollution due to automobiles and control methods. (10 Marks)
b. State various environmental laws and acts on Air pollution. (10 Marks)

OR

- 10 a. Explain Noise pollution causes, effects and control measures. (10 Marks)
b. Explain briefly global Episodes :
i) London smog
ii) Bhopal gas tragedy. (10 Marks)

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18CV733

Seventh Semester B.E. Degree Examination, Dec.2024/Jan.2025 Pavement Materials and Construction

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain in detail the desirable properties of road aggregate. (10 Marks)
- b. Explain Rothfutch's method for gradation of aggregates with the aid of graph. (10 Marks)

OR

- 2 a. Differentiate between Bitumen and Tar. (05 Marks)
- b. Explain the manufacturing process of Tar. List the different grades of tar with its application and also the tests conducted on road tar. (10 Marks)
- c. List the requirements and desirable properties of bitumen. (05 Marks)

Module-2

- 3 a. Explain Bituminous Emulsion? Explain the properties of Road emulsion on different tests. (08 Marks)
- b. Explain different types of Bituminous Emulsion. (04 Marks)
- c. Explain the methods of improving Adhesion. (08 Marks)

OR

- 4 a. Write short notes on cutback bitumen and types of cutback bitumen. (08 Marks)
- b. List the tests conducted on cutback bitumen. (04 Marks)
- c. Mention the types of adhesion test. Explain in brief Static Immersion test. (08 Marks)

Module-3

- 5 a. Explain the desirable properties of Bituminous mixes. (08 Marks)
- b. Explain the procedure of determining the optimum bitumen content for a bituminous mix by Marshall test method. (12 Marks)

OR

- 6 a. List the steps followed for the design of bituminous mix. (08 Marks)
- b. The specific gravities and weight proportion for aggregates and bitumen are as under the preparation of Marshall mix design. The volume and weight of one Marshall specimen was found to be 475 cc and 1100 gm. Assume absorption of bitumen in aggregate is zero. Find V_v , V_b , VMA and VFB.

Item	A-1	A-2	A-3	A-4	Bitumen
Weight (gm)	825	1200	325	150	100
Specific gravity	2.63	2.51	2.46	2.43	1.05

(12 Marks)

Module-4

- 7 a. Discuss the need for construction of embankment. What are the important elements considered in embankment design? (08 Marks)
- b. Enumerate the steps in the construction of subgrade. What are the quality control checks for subgrade construction. (12 Marks)

OR

- 8 a. Briefly explain different types of rollers used in road construction with its suitability. (10 Marks)
- b. Explain working principle of power shovel with a neat sketch. List its applications and limitations. (10 Marks)

Module-5

- 9 a. Explain the material specification and construction steps for Bituminous Macadam. (12 Marks)
- b. Write short notes on : (08 Marks)
- i) Prime coat
 - ii) Tackcoat.

OR

- 10 a. Mention the advantages of providing Dry Lean Concrete (DLC) as subbase. (08 Marks)
- b. Explain expansion joint and contraction joint. (08 Marks)
- c. List the quality control test conducted on concrete road pavement. (04 Marks)

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18CV745

Seventh Semester B.E. Degree Examination, Dec.2024/Jan.2025 Urban Transport Planning

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What is urban transportation planning? List the steps in the process of transportation planning and explain them briefly. (10 Marks)
- b. What is co-ordination of transport systems? List and explain the types of co-ordination. (10 Marks)

OR

- 2 a. Explain the systems approach in transport planning with a flow chart. (10 Marks)
- b. Define urbanization. Explain the effects of urbanization in transportation sector. (10 Marks)

Module-2

- 3 a. What is Zoning? List out the points to be considered during zoning. (10 Marks)
- b. Define sampling. List and explain briefly the types of sampling. (10 Marks)

OR

- 4 a. Explain the process of road side interviews conducted for collecting the data for transport planning. (10 Marks)
- b. Define external cordon line. List the types of inventories conducted for finding transport facilities, explain briefly. (10 Marks)

Module-3

- 5 a. What is trip generation? List and explain the factors affecting. Trip generation and attraction rates. (10 Marks)
- b. The trip rate 'y' and the corresponding household sizes 'x' from a sample are shown in the table below. Compute the trip rate. If the avg household size is 3.25. Establish the trip generation equation.

Household size (x) →				
Zones →	1	2	3	4
Trips per day (y) ↓	1	3	4	5
	3	4	5	8
	3	5	7	8

(10 Marks)

OR

- 6 a. What are the assumptions made in multilinear regression analysis? List and explain the types of multilinear regression analysis. (10 Marks)
- b. The following data shows average household. Size and total trips made per day for a particular zone of study area. Develop the trip production equation and also compute coefficient of correlation.

Average Household size (x)	Total trips/day (y)
2	4
3	6
4	7
5	8
6	10

(10 Marks)

Module-4

- 7 a. What is Modal split? Explain with a flow chart, the modal split carried out after trip distribution. (10 Marks)
- b. Obtain the future trip table by average factor method given the expected future trips for zone 1, 2, 3 are expected to be 360, 1260 and 3120 respectively.

D \ O	1	2	3
1	60	100	200
2	100	20	300
3	200	300	20

(10 Marks)

OR

- 8 a. List and explain the factors affecting the modal split. (10 Marks)
- b. The total trips produced and attracted to the 3 zones A, B, C of a survey area in the design years are tabulated as follows:

Zone	Trips produced	Trips attracted
A	2000	3000
B	3000	4000
C	4000	2000

It is known that the trips between 2 zones are inversely proportional to the second power of the travel time between zones, which is uniformly 20 min. If the trip interchange between B and C is known to be 600. Calculate the trip interchange between zones A and B, A and C, B and A, C and A, C and B.

(10 Marks)

Module-5

- 9 a. Define Traffic Assignment. List the various methods of route assignment. Explain any two methods. (10 Marks)
- b. With a flow chart, explain the structure of a lowry model. (10 Marks)

OR

- 10 a. Explain the major requirements of the traffic assignment. (10 Marks)
- b. What are the major factors to be considered while selecting a land-use model? Explain briefly. (10 Marks)

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18CV753

Seventh Semester B.E. Degree Examination, Dec.2024/Jan.2025 Environmental Protection and Management

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What are the unique characteristics of environmental problems faced by mankind today? Discuss some of the critical environmental issues experienced on local, regional and global scale and their probable mitigation measures. (08 Marks)
- b. Define "Environmental Stewardship". How does the local environmental stewardships action have an impact on the socio-ecological outcomes of the region? (04 Marks)
- c. Distinguish between:
 - i) Polluter pays principle and user pays principle.
 - ii) Principle of participation and principle of proportionality. (08 Marks)

OR

- 2 a. Explain briefly the national policies on environment abatement of pollution and conservation of natural resources and the relevant legislations enacted by the Indian government to safeguard the environment. (10 Marks)
- b. Write a note on "Business charter for sustainable production and consumption". (10 Marks)

Module-2

- 3 a. Distinguish between:
 - i) Stream standards and effluent standards
 - ii) Emission and ambient standards. (12 Marks)
- b. What are the objectives of conducting an environmental performance evaluation of an organization and mention the benefits derived from it. (08 Marks)

OR

- 4 a. Distinguish between:
 - i) Pollution control and pollution prevention
 - ii) Clean technologies and zero discharge technologies. (12 Marks)
- b. Write a note on:
 - i) Environmental performance indicators and
 - ii) Bench marking. (08 Marks)

Module-3

- 5 a. Explain the basic principles of ISO:14001. (10 Marks)
- b. How does initial environmental review help in creating Environmental Management System (EMS) of an organization. (10 Marks)

OR

- 6 a. What is EMS? What are the basic elements of EMS? (10 Marks)
- b. Discuss the benefits derived from EMS. (10 Marks)

Module-4

- 7 a. How “Environmental Audit (EA)” differs from “Environmental Impact Assessment (EIA)”?
Explain. (10 Marks)
b. Write a note on ISO:19001-2018 auditing management systems. (10 Marks)

OR

- 8 a. Discuss various types of environmental audits. (10 Marks)
b. What are the roles and responsibilities of auditors conducting an environmental audit? (05 Marks)
c. Write a note on “Waste Audit”. (05 Marks)

Module-5

- 9 a. Discuss the application of EMS and pollution prevention in electroplating and tanning industry. (14 Marks)
b. What are the four characteristics which defines a hazardous waste? What are the various minimization techniques and disposal methods adopted in managing hazardous waste? (06 Marks)

OR

- 10 a. Discuss the application of EMS and pollution prevention in sugar and textile industry. (14 Marks)
b. Write a note on transboundary movement of hazardous waste. (06 Marks)

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Third Semester B.E. Degree Examination, Dec.2024/Jan.2025 Transform Calculus, Fourier Series and Numerical Techniques

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Find $L\left\{\frac{\cos at - \cos bt}{t}\right\}$. (06 Marks)
- b. A periodic function of period $\frac{2\pi}{\omega}$ is defined by, $f(t) = \begin{cases} E \sin \omega t, & 0 < t < \frac{\pi}{\omega} \\ 0, & \frac{\pi}{\omega} < t < \frac{2\pi}{\omega} \end{cases}$ (07 Marks)
- c. Solve : $y'' - y' - 2y = 0$; given $y(0) = 0$ and $y'(0) = 6$ by Laplace transformation method. (07 Marks)

OR

- 2 a. Find $L^{-1}\left\{\frac{4s+5}{(s-1)^2(s+2)}\right\}$. (06 Marks)
- b. Apply convolution theorem to evaluate $L^{-1}\left\{\frac{1}{(s^2+1)(s^2+9)}\right\}$. (07 Marks)
- c. Using unit step function, find the Laplaces transform of, $f(t) = \begin{cases} \sin t, & 0 \leq t < \pi \\ \sin 2t, & \pi \leq t \leq 2\pi \\ \sin 3t, & t \geq 2\pi \end{cases}$. (07 Marks)

Module-2

- 3 a. Obtain the Fourier Series for the function, $f(x) = x^2$, $0 < x < 2\pi$. (06 Marks)
- b. Find the Fourier series of $f(x)$,
Where $f(x) = \begin{cases} -\pi, & -\pi < x < 0 \\ x, & 0 < x < \pi \end{cases}$. (07 Marks)
- c. Express $f(x) = x$ as a half-range cosine series in $0 < x < 2$. (07 Marks)

OR

- 4 a. Obtain Fourier series for the function, $f(x) = \begin{cases} \pi x, & 0 \leq x \leq 1 \\ \pi(2-x), & 1 \leq x \leq 2 \end{cases}$. (06 Marks)
- b. Find the Fourier half-range cosine series of the function $f(x) = (x+1)$, in $(0, 1)$. (06 Marks)
- c. Compute the first harmonic of the Fourier series of $f(x)$ given in the following table :

x	0	$\frac{\pi}{3}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{5\pi}{3}$	2π
y	1.0	1.4	1.9	1.7	1.5	1.2	1.0

(08 Marks)

Module-3

- 5 a. Find the z-transform of, $3n - 4\sin\left(\frac{n\pi}{4}\right) + 5a$. (06 Marks)
- b. Compute the inverse z – transform of, $\frac{2z^2 + 3z}{(z+2)(z-4)}$. (07 Marks)
- c. Find the Fourier transform of, $f(x) = \begin{cases} 1, & |x| < 1 \\ 0, & |x| > 1 \end{cases}$, Hence evaluate $\int_0^{\infty} \frac{\sin x}{x} dx$. (07 Marks)

OR

- 6 a. Find the Fourier sine transform of e^{-ax} . (06 Marks)
- b. If $U(z) = \frac{2z^2 + 5z + 14}{(z-1)^4}$, evaluate u_2 and u_3 . (07 Marks)
- c. Using the z-transform, solve $u_{n+2} + 4u_{n+1} + 3u_n = 3^n$, $u_0 = 0$, $u_1 = 1$. (07 Marks)

Module-4

- 7 a. Find by Taylor's series method the value of y at $x = 0.1$ and $x = 0.2$ to four places of decimals from, $\frac{dy}{dx} = x^2y - 1$, $y(0) = 1$. (06 Marks)
- b. Apply Runge-Kutta fourth order method to find an approximate value of y when $x = 0.2$, given $\frac{dy}{dx} = x + y$, $y(0) = 1$. (07 Marks)
- c. If $\frac{dy}{dx} = 2e^x - y$, $y(0) = 2$, $y(0.1) = 2.010$, $y(0.2) = 2.04$ and $y(0.3) = 2.09$, find $y(0.4)$ by employing the Milne's predictor-corrector formula, use corrector formula twice. (07 Marks)

OR

- 8 a. Using modified Euler's method, solve the IVP $\frac{dy}{dx} = x + \sqrt{y}$, $y(0) = 1$ at $x = 0.2$, perform three modifications. (06 Marks)
- b. Using the fourth order Runge-Kutta method, solve the IVP $\frac{dy}{dx} = \frac{1}{x+y}$ at the point $x = 0.5$. Given that $y(0.4) = 1$. (07 Marks)
- c. Given $\frac{dy}{dx} = x^2(1+y)$, $y(1) = 1$, $y(1.1) = 1.233$, $y(1.2) = 1.548$, $y(1.3) = 1.979$. Determine $y(1.4)$ by Adams-Bashforth method. (07 Marks)

Module-5

- 9 a. Using Runge-Kutta method of fourth order solve the differential equation, $\frac{d^2y}{dx^2} = x\left(\frac{dy}{dx}\right)^2 - y^2$, with $y(0) = 1$, $y'(0) = 0$ at $x = 0.2$. (06 Marks)
- b. Derive Euler's equation in the standard form, $\frac{\partial f}{\partial y} - \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) = 0$. (07 Marks)

- c. On which curve the functional,

$$\int_0^{\frac{\pi}{2}} (y'^2 - y^2 + 2xy) dy, y(0) = 0, y\left(\frac{\pi}{2}\right) = 0 \text{ be extremized.}$$

(07 Marks)

OR

- 10 a. Apply Milne's method to compute $y(0.8)$, given that $\frac{d^2y}{dx^2} = 1 - 2y \frac{dy}{dx}$ and

x	0	0.2	0.4	0.6
y	0	0.02	0.0795	0.1762
y'	0	0.1996	0.3937	0.5689

(06 Marks)

- b. Prove that the geodesics on a plane are straight line.

(07 Marks)

- c. Find the extremal of the functional, $I = \int_0^{\frac{\pi}{2}} (y'^2 - y^2 + 4y \cos x) dx$, given that $y(0) = 0$,

$$y\left(\frac{\pi}{2}\right) = 0.$$

(07 Marks)

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Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Advanced Mathematical Methods

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Reduce the matrix $A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 1 & 2 \\ 4 & 2 & 1 \end{bmatrix}$ to tridiagonal form and find its eigen values using

Given's method

(12 Marks)

- b. Find the eigen values of the following matrix by House holder method $A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & 2 \\ -1 & 2 & 1 \end{bmatrix}$

(08 Marks)

OR

- 2 a. Use the Birge Vieta method to find a real root correct upto three decimal places for the equation $x^3 - 11x^2 + 32x - 22 = 0$. Take $P_0 = 0.5$. (12 Marks)
- b. Using 'Bairstow' method, find the quadratic factor of $x^3 + x^2 - x + 2 = 0$ by taking $P = -0.9$, $q = 0.9$. (08 Marks)

Module-2

- 3 a. Solve $u_{xx} + u_{yy} = 0$ for the following square with the boundary conditions as indicated in the Fig Q3(a).

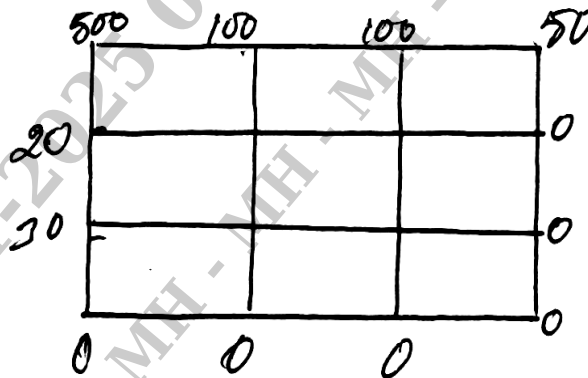


Fig Q3(a)

(10 Marks)

- b. Solve the Poisson equation $u_{xx} + u_{yy} = 8x^2y^2$ for the following square mesh with $u(x,y) = 0$ on the boundary and mesh length equal to 1.

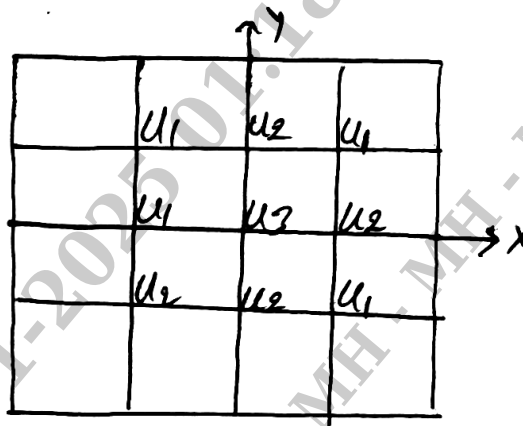


Fig Q3(b)

(10 Marks)

OR

- 4 a. Solve the wave equation $u_{tt} = 4u_{xx}$ subjected to $u(0, t) = 0$, $u(4t) = 0$, $u_t(x, 0) = 0$ and $u(x, 0) = x(4 - x)$ by taking $h = 1$, $k = 0.5$. (10 Marks)
- b. Using Crank – Nicholson method, solve $u_{xx} = 16 u_t$, $0 < x < 1$, $t > 0$, given $u(x, 0) = 0$, $u(0, t) = 50t$. Compute u for two steps in the direction of t by taking $h = 0.25$. (10 Marks)

Module-3

- 5 a. Find the Taylor's series expansion of $f(z) = \frac{1}{z^2 + 4}$ about $z = -i$. (07 Marks)

b. Evaluate $\int_0^{2\pi} \frac{d\theta}{5 - 3\cos\theta}$. (07 Marks)

c. Find the residue of $f(z) = \frac{ze^2}{(z-a)^3}$ at its pole. (06 Marks)

OR

- 6 a. Find the Laurent's series expansion of $f(z) = \frac{1}{(z+1)(z+3)}$ in $1 < |z| < 3$. (07 Marks)

b. Using Cauchy's residue theorem, evaluate $\int_C \frac{(4-3z)dz}{z(z-1)(z-2)}$, where C is the circle $|z| = \frac{3}{2}$. (07 Marks)

c. Evaluate $\int_0^\infty \frac{dx}{(1+x^2)^3}$, using contour integration. (06 Marks)

Module-4

- 7 a. A coin is tossed 900 times and head appears 490 times, can it be concluded that the coin is biased at 5% level of significance. Given $z_{0.05} = 1.96$. (07 Marks)
- b. An examination was conducted to two groups of students of two colleges. The following is the given data :
 College A : $\mu_A = 75, \sigma_A = 9, n_1 = 90$
 College B : $\mu_B = 79, \sigma_B = 7, n_2 = 60$
 Is there a significant difference in the performance of two colleges at 5% level of significance. Given $z_{0.05} = 1.96$. (07 Marks)
- c. The following table represent the production of three varieties of crops A, B, C, each grown on 4 plots. Find whether the variety differences are significant. Given $F_c = 4.26$.

Production/Acre			
Plot	A	B	C
1	6	5	5
2	7	5	4
3	3	3	-3
4	8	7	4

(06 Marks)

OR

- 8 a. A die is thrown 9000 times. The numbers 3 or 4 appears 3240 times. Test whether the die is unbiased at 5% level of significance. Given that $z_{0.05} = 1.96$. (07 Marks)
- b. Two independent samples of sizes 7 and 6 have the following data :
 Sample A : 28 30 32 33 31 29 34
 Sample B : 29 30 30 24 27 28 -
 Examine whether the samples have been drawn from a normal population having the same variance. Given $F_{0.05} = 4.39$. (07 Marks)
- c. Write a short note on :
 i) Null hypothesis ii) Types of errors iii) level of significance. (06 Marks)

Module-5

- 9 a. Show that the process $x(t) = A \cos (wt + \theta)$ is wide sense stationary. Where A and w are constants and θ is uniformly distributed random variable in $(0, 2\pi)$. (07 Marks)

- b. Find the fixed probability vector of the regular Stochastic matrix $A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{pmatrix}$. (07 Marks)
- c. In a railway yard, trains arrive at a rate of 30 trains per day. Assuming that, the inter arrival time and service time distribution are exponential with an average of 36 minutes. Find :
 i) The probability that the yard is empty ii) average queue length, if the line capacity of the yard is 9 trains. (06 Marks)

OR

- 10 a. Three boy A, B and C are throwing a ball to each other. A always throws the ball to B and B always throws the ball to C, but C is just likely to throw the ball to B as to A. If C is the first to throw the ball, find the probabilities that for the fourth throw
i) A has the ball ii) B has the ball iii) C has the ball. (07 Marks)
- b. Cars arrive at a petrol pump, having one petrol unit, in a Poisson Fashion, with an average of 10 cars per hour. The service time is exponentially distributed with a mean of 3 minutes. Find : i) average number of cars in the system ii) average queue length iii) probability that the number of cars in the system is 2. (07 Marks)
- c. The auto correlation function for a stationary process is given by $R_{xx}(\tau) = 25 + \frac{4}{1+6\tau^2}$.
Find the mean and variance of the process $X(t)$. (06 Marks)

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